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University

Project Title: Creating an Integrated Database Solution for Global Computer Solutions (GCS): Streamlining Project Management, Employee Assignment, and the Billing Processes.

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1. Critical Evaluation of the Database System

1.1 Introduction

Strong database systems are essential to the seamless operation of businesses like Global Computer Solutions (GCS) in the fast-paced world of technology. These databases are the backbone of managing everything from projects to employee data and billing processes. For GCS, selecting the right type of database is vital to ensure their systems can handle growth, maintain performance, and keep data secure (Laudon et al, 2020). In this evaluation, we'll look at different types of databases, considering factors like how well they can grow as GCS expands, how fast they can process data, and how reliable they are in keeping GCS's information safe. By examining these aspects, we aim to recommend the best database solution for GCS's requirements.

1.2 Factors to consider while choosing the Database Technology

To critically evaluate existing database technologies and suggest the best option for a software company like Global Computer Solutions (GCS), we need to consider several factors:

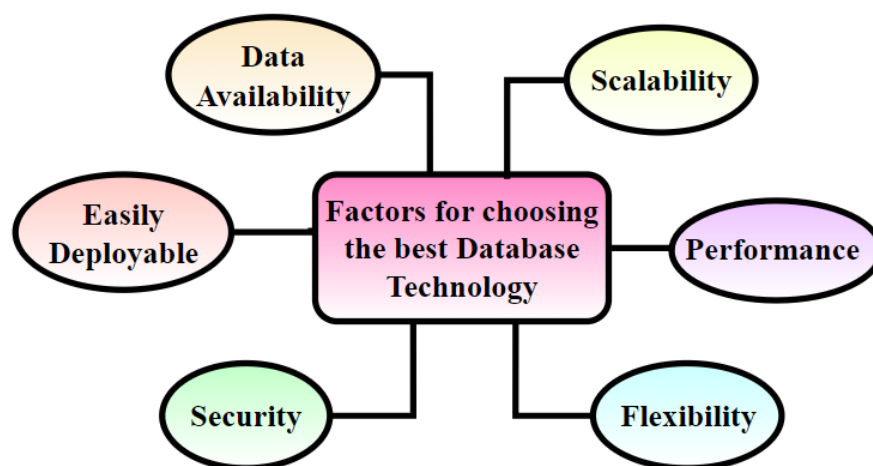


Figure 1 Factors for best database technology

1. Scalability is useful to handle the increased amount of data and user loads. (Kleppmann, M., 2017)
2. High performance from the database is required to enable real-time transactions. (Schwartz, 2012)
3. The system should be sufficiently dependable to offer increased data availability, lower downtime, and minimize losses to the business and work interruptions. (Connolly, T., & Begg, C., 2014)
4. The changing business requirements require flexibility to change the complete business models that are stored in a database. (Ralph Kimball et al, 2013)
5. Strong security measures should be provided by the database of choice to shield data from hacker attacks, illegal access, and other risks. (Afyouni, H.A., 2006)
6. The database should be easy to deploy, manage, and maintain, with tools for monitoring, backup, and recovery as well as provide integration with other technologies. (Mullins, C., 2002)

1.3 Types of Database Technologies

1. Hierarchical databases

The organization of data in a hierarchical database model resembles a tree. Each field in the collection of fields that stores the data has a single value in it. Through connections, the records are connected to the parent-child relationship. Every child record in a hierarchical database model has a single parent. A parent may have more than one child. The tree is traversed until the record is found to retrieve the data. IBM Information Management System (IMS) and Windows Registry are famous examples of hierarchical databases. (Navathe et al, 2017)

Advantage: It is quick to access and change a hierarchical database.

Disadvantage: Since there is only one parent per child in the tree, links or ties between children are not allowed. (Sudarshan et al, 2020)

2. Relational databases

In relational databases, data relationships are relational and are stored in tabular form using columns and rows in relational database management systems (RDBMS). Every table row denotes a record, and every table column denotes an attribute. A table's fields each correspond to a data value. It also has ACID properties. PostgreSQL, MySQL, Oracle, and Microsoft SQL Server are some examples of relational databases. (Silberschatz et al, 2020)

Advantage: Relational databases require minimal to no training to use and also entries can be modified without specifying the entire body.

Disadvantage: It has limited scalability.
(Silberschatz et al, 2020)

3. NoSQL databases

In NoSQL databases we do not have predefined schemas, they're an excellent choice for development environments that change quickly. It allows changes quickly without impacting the applications. Column, Document, Graph, Key-value, and Object are the five main categories into which NoSQL databases fall. MongoDB, Cassandra, and Redis Server are some examples of non-relational databases. (Sadalage et al, 2012)

Advantage: Relational databases require minimal to no training to use and also entries can be modified without specifying the entire body.

Disadvantage: Do not support ACID properties like the relational databases.
(Sadalage et al, 2012)

4. Cloud databases

In Cloud database information is processed using cloud computing and stored in a virtual environment. A variety of cloud computing services (SaaS - Software as a Service, PaaS - Platform as a Service, IaaS - Infrastructure as a Service, etc.) are available to customers with database access. Some cloud databases are AWS, Microsoft Azure, and Google Cloud SQL. (Mahmood et al, 2013)

Advantage: It is highly scalable and accessible as well as useful for recovering data during natural disasters.

Disadvantage: System downtime may create issues for users but by using clusters this may be resolved.

(Rajkumar Buyya and S. N. S, 2019)

1.4 Best Database Technology for the Case Study

Relational database technology is best suited for the case study for several reasons as mentioned below:

1. The case study has highly organized data, including work logs, client information, project details, and staff information. RDBMS is nice to choose for handling structured data with clear schemas. (Elmasri et al, 2016)
2. Relational database management systems (RDBMSs) make sure that transactions are handled with extreme care, ensuring they follow the ACID principles: Atomicity, Consistency, Isolation, and Durability. This means that every transaction is processed reliably from start to finish, maintaining the integrity of your data. (Elmasri et al, 2016)
3. RDBMSs can set constraints on data integrity, including those related to uniqueness, referential integrity, and data validation standards. (Elmasri et al, 2016)
4. The relationships between different entities in the case study, such as employees assigned to projects, tasks associated with projects, and work logs tied to specific assignments, can be efficiently represented using the relational model. This model allows for defining and enforcing relationships between tables through foreign key constraints. (Elmasri et al, 2016)

Conclusion: The greatest option for handling the data and operations of Global Computer Solutions (GCS) would be a well-designed relational database made with software such as MySQL, PostgreSQL, or Microsoft SQL Server. Complex connections between projects, tasks, customers, and employees require the support, structure, and data integrity that relational databases offer. Relational databases also offer robust security features, which are essential for securing both client and project information. (Valdurez, P., 2011)

While cloud databases and NoSQL databases offer advantages in scalability and flexibility (Redmond, E., 2018), a relational database aligns well with the structured nature of the data and the relational dependencies described in the case study.

2. Database Design

2.1 Entity Relationship Diagram (ERD) (Logical Model)

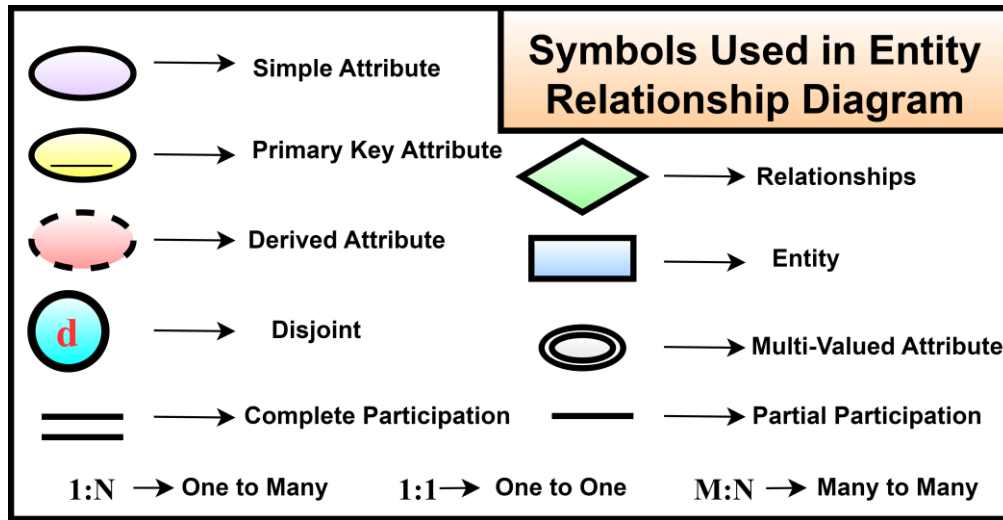


Figure 2 Symbols Used in E-R Diagram

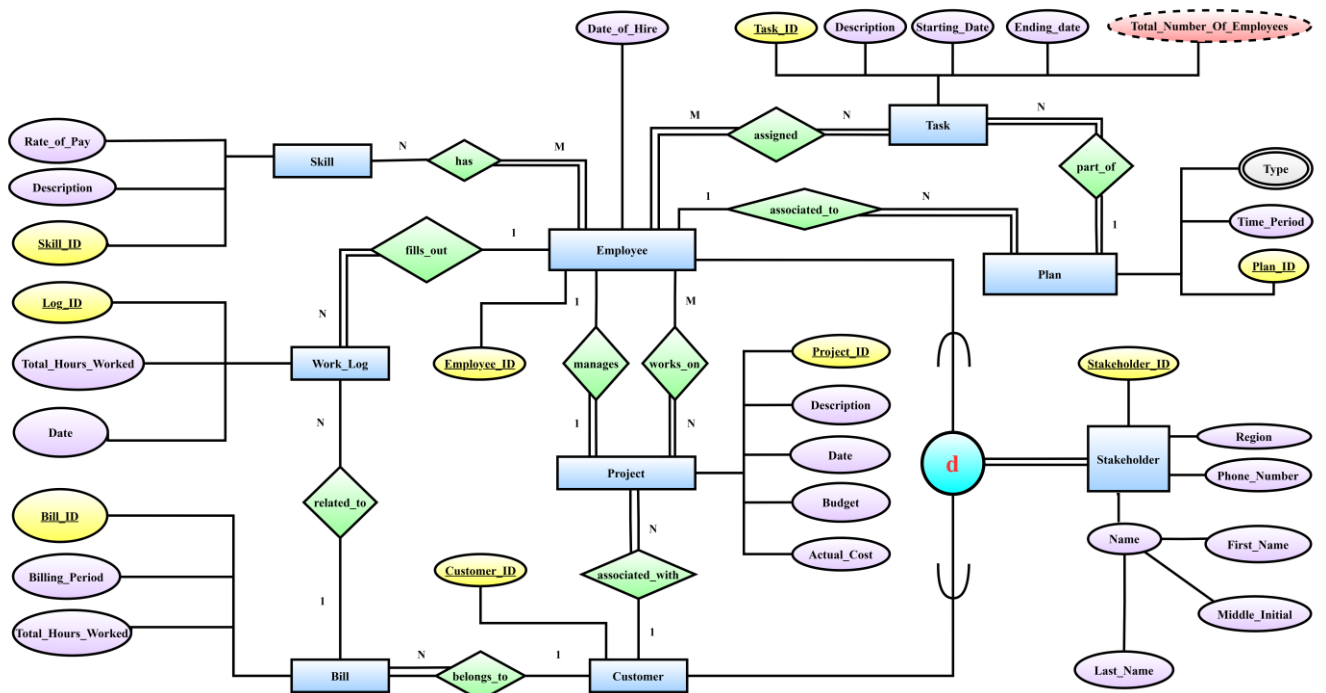


Figure 3 Entity Relationship Diagram (Logical Model) for Global Computer Solutions (GCS)

2.2 Relational Schema Mapping

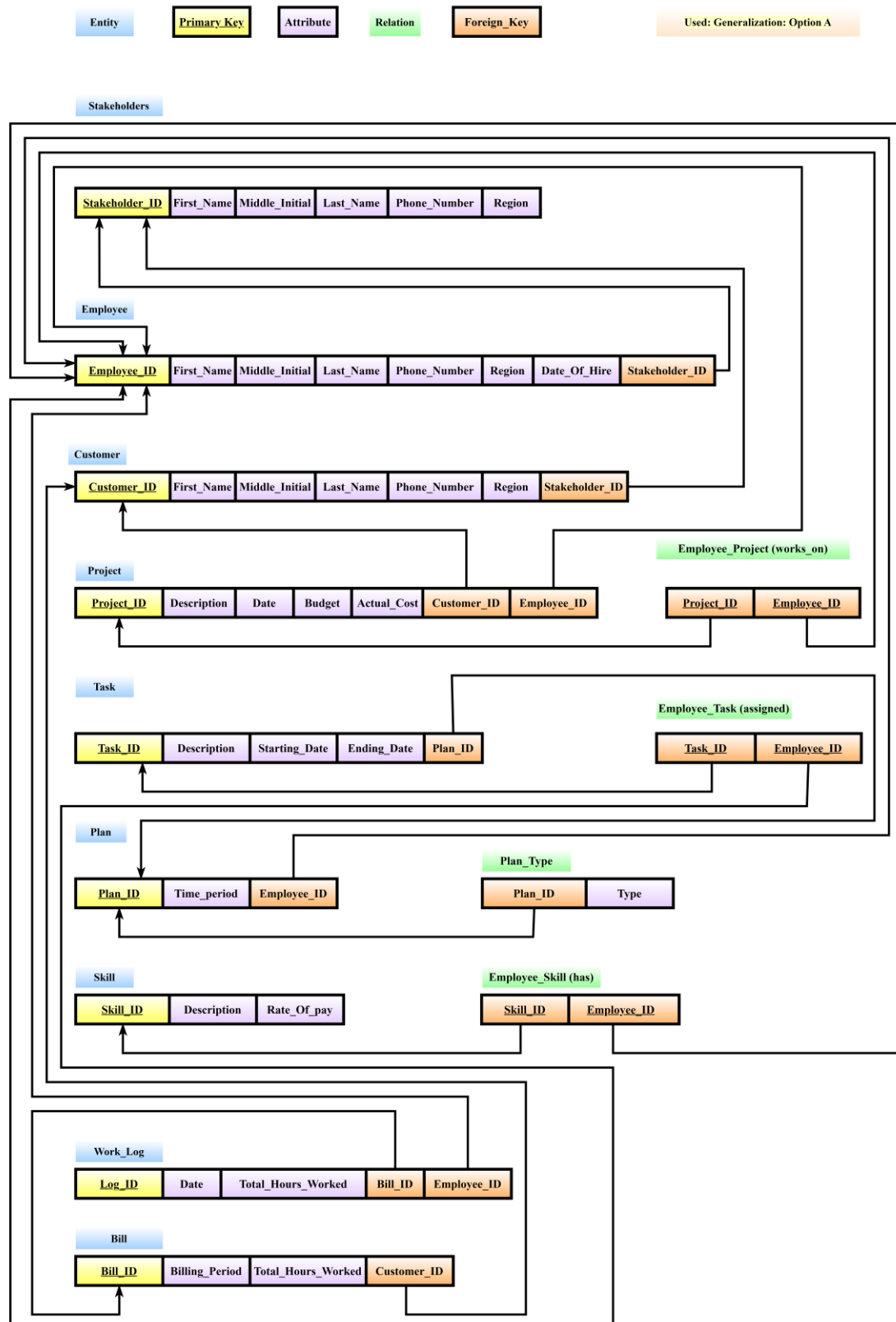


Figure 4 Relational Schema Mapping for Global Computer Solutions (GCS)

2.3 Assumptions and Changes

- The superclass named “Stakeholders” is an assumption to have common attributes like name, region, and phone_number (provided in the case study for the customer but assumed for the employee) comprised of two sub-classes already named “Customer” and “Employee” given in the case study.
- The manager manages one project at a time (given in the case study) whereas each project has many employees working on it.
- A task requires one or many skills to complete, and it is assumed that employees assigned to the task already have those skills.
- A customer can only view the project and the bill generated for the project.
- The Manager is responsible for the execution of the plan related to the task assigned to an employee.
- Generalization having option – (A) has been used in the designing of this database system for Global Computer Solutions (GCS) including the schema.

3. Database Development

3.1 Physical Design of the System

The physical design of the system, screenshot for the entire system from the MySQL server (Design Tab)

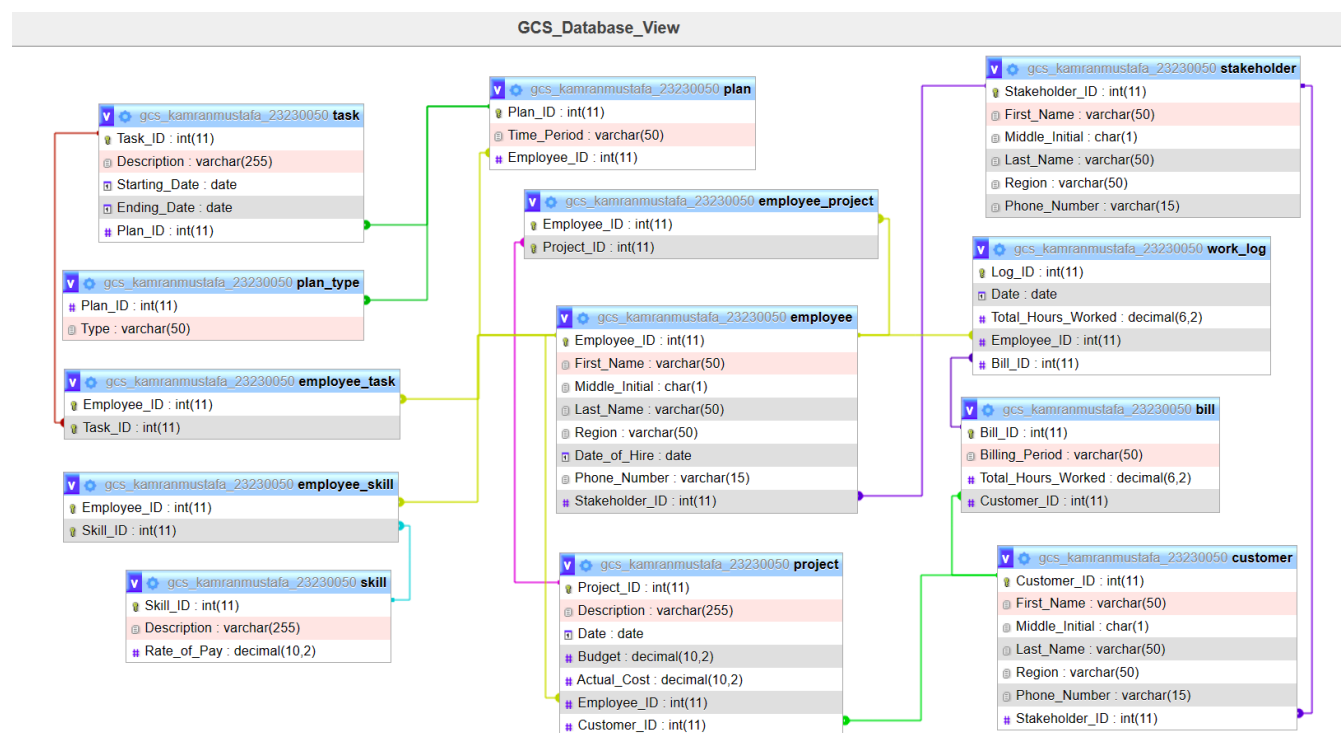


Figure 5 Global Computer Solutions (GCS) physical Database Schema and Relationship Representation

3.2 Tables and Relations

This is an overview of all entities with their relations available in the Global Computer System database.

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050

Structure SQL Search Query Export Import Operations Privileges Routines Events Tr

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ bill	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ customer	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ employee	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ employee_project	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ employee_skill	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ employee_task	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ plan	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ plan_type	★ Browse Structure Search Insert Empty Drop	40	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ project	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	48.0 KiB	-
☐ skill	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	16.0 KiB	-
☐ stakeholder	★ Browse Structure Search Insert Empty Drop	30	InnoDB	latin1_swedish_ci	16.0 KiB	-
☐ task	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	32.0 KiB	-
☐ work_log	★ Browse Structure Search Insert Empty Drop	15	InnoDB	latin1_swedish_ci	48.0 KiB	-
13 tables	Sum	235	InnoDB	latin1_swedish_ci	416.0 KiB	0 B

⬆️ ☐ Check all With selected:

Figure 6 Total Tables (Entities) for Global Computer Solutions (GCS)

3.2.1 Stakeholder

The Stakeholder table stores the data of stakeholders (customers and employees) with a primary key as Stakeholder_ID and has other details such as First_Name, Middle_Initial, Last_Name, Region, and Phone_Number. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	Stakeholder_ID	int(11)			No	None			Change Drop More
☐ 2	First_Name	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐ 3	Middle_Initial	char(1)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐ 4	Last_Name	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐ 5	Region	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐ 6	Phone_Number	varchar(15)	latin1_swedish_ci		Yes	NULL			Change Drop More

Figure 7 Stakeholder Entity Structure

+ Options

				Stakeholder_ID	First_Name	Middle_Initial	Last_Name	Region	Phone_Number
☐				10001	Robert	A	Wilson	North America	555-1234
☐				10002	Jennifer	B	Anderson	Europe	555-5678
☐				10003	Christopher	C	Thompson	Asia	555-9012
☐				10004	Jessica	D	Harris	Australia	555-3456
☐				10005	Matthew	E	Taylor	Africa	555-7890
☐				10006	Amanda	F	Moore	South America	555-2345
☐				10007	David	G	Jackson	Europe	555-6789
☐				10008	Sarah	H	White	North America	555-0123
☐				10009	Daniel	I	Adams	Asia	555-4567
☐				10010	Ashley	J	Nelson	Australia	555-8901
☐				10011	Michael	K	Roberts	Africa	555-2345
☐				10012	Emily	L	Thomas	South America	555-6789
☐				10013	Christopher	M	Hill	Europe	555-1234
☐				10014	Michelle	N	Young	North America	555-5678
☐				10015	Joseph	O	King	Asia	555-9012
☐				10016	Maria	A	Gonzalez	North America	555-1234
☐				10017	David	B	Perez	Europe	555-5678
☐				10018	Mary	C	Sanchez	Asia	555-9012
☐				10019	Carlos	D	Rivera	Australia	555-3456
☐				10020	Laura	E	Gomez	Africa	555-7890
☐				10021	Pedro	F	Diaz	South America	555-2345
☐				10022	Ana	G	Torres	Europe	555-6789

				Stakeholder_ID	First_Name	Middle_Initial	Last_Name	Region	Phone_Number
☐				10022	Ana	G	Torres	Europe	555-6789
☐				10023	Juan	H	Vargas	North America	555-0123
☐				10024	Luis	I	Lopez	Asia	555-4567
☐				10025	Miguel	J	Martinez	Australia	555-8901
☐				10026	Elena	K	Garcia	Africa	555-2345
☐				10027	Rosa	L	Hernandez	South America	555-6789
☐				10028	Javier	M	Fernandez	Europe	555-1234
☐				10029	Gloria	N	Moreno	North America	555-5678
☐				10030	Sofia	O	Alvarez	Asia	555-9012

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Figure 8 Stakeholder Entity Data Entries

3.2.2 Employee

The Employee table stores the data of employees with a primary key as Employee_ID and has other details such as First_Name, Middle_Initial, Last_Name, Region, Date_Of_Hire, and Phone_Number. Stakeholder_ID in this table is a foreign key referencing from the stakeholder table. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 Employee_ID	int(11)			No	None			Change Drop More
☐	2 First_Name	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	3 Middle_Initial	char(1)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	4 Last_Name	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	5 Region	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	6 Date_of_Hire	date			Yes	NULL			Change Drop More
☐	7 Phone_Number	varchar(15)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	8 Stakeholder_ID	int(11)			Yes	NULL			Change Drop More

Figure 9 Employee Entity Structure

+ Options

			Employee_ID	First_Name	Middle_Initial	Last_Name	Region	Date_of_Hire	Phone_Number	Stakeholder_ID	
				20001	Robert	A	Wilson	North America	2023-03-15	555-1234	10001
				20002	Jennifer	B	Anderson	Europe	2023-04-20	555-5678	10002
				20003	Christopher	C	Thompson	Asia	2023-05-10	555-9012	10003
				20004	Jessica	D	Harris	Australia	2023-06-05	555-3456	10004
				20005	Matthew	E	Taylor	Africa	2023-07-12	555-7890	10005
				20006	Amanda	F	Moore	South America	2023-08-25	555-2345	10006
				20007	David	G	Jackson	Europe	2023-09-30	555-6789	10007
				20008	Sarah	H	White	North America	2023-10-17	555-0123	10008
				20009	Daniel	I	Adams	Asia	2023-11-22	555-4567	10009
				20010	Ashley	J	Nelson	Australia	2023-12-01	555-8901	10010
				20011	Michael	K	Roberts	Africa	2024-01-05	555-2345	10011
				20012	Emily	L	Thomas	South America	2024-02-10	555-6789	10012
				20013	Christopher	M	Hill	Europe	2024-03-20	555-1234	10013
				20014	Michelle	N	Young	North America	2024-04-15	555-5678	10014
				20015	Joseph	O	King	Asia	2024-05-05	555-9012	10015
					</						

Figure 10 Employee Entity Data Entries

3.2.3 Customer

The Customer table stores the data of customers with a primary key as Customer_ID and has other details such as First_Name, Middle_Initial, Last_Name, Region, and Phone_Number. Stakeholder_ID in this table is a foreign key referencing from the stakeholder table. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 Customer_ID	int(11)			No	None			Change Drop More
☐	2 First_Name	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	3 Middle_Initial	char(1)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	4 Last_Name	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	5 Region	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	6 Phone_Number	varchar(15)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	7 Stakeholder_ID	int(11)			Yes	NULL			Change Drop More

Figure 11 Customer Entity Structure

+ Options

			Customer_ID	First_Name	Middle_Initial	Last_Name	Region	Phone_Number	Stakeholder_ID	
☐	 Edit	 Copy	 Delete	30001	Maria	A	Gonzalez	North America	555-1234	10016
☐	 Edit	 Copy	 Delete	30002	David	B	Perez	Europe	555-5678	10017
☐	 Edit	 Copy	 Delete	30003	Mary	C	Sanchez	Asia	555-9012	10018
☐	 Edit	 Copy	 Delete	30004	Carlos	D	Rivera	Australia	555-3456	10019
☐	 Edit	 Copy	 Delete	30005	Laura	E	Gomez	Africa	555-7890	10020
☐	 Edit	 Copy	 Delete	30006	Pedro	F	Diaz	South America	555-2345	10021
☐	 Edit	 Copy	 Delete	30007	Ana	G	Torres	Europe	555-6789	10022
☐	 Edit	 Copy	 Delete	30008	Juan	H	Vargas	North America	555-0123	10023
☐	 Edit	 Copy	 Delete	30009	Luis	I	Lopez	Asia	555-4567	10024
☐	 Edit	 Copy	 Delete	30010	Miguel	J	Martinez	Australia	555-8901	10025
☐	 Edit	 Copy	 Delete	30011	Elena	K	Garcia	Africa	555-2345	10026
☐	 Edit	 Copy	 Delete	30012	Rosa	L	Hernandez	South America	555-6789	10027
☐	 Edit	 Copy	 Delete	30013	Javier	M	Fernandez	Europe	555-1234	10028
☐	 Edit	 Copy	 Delete	30014	Gloria	N	Moreno	North America	555-5678	10029
☐	 Edit	 Copy	 Delete	30015	Sofia	O	Alvarez	Asia	555-9012	10030

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Figure 12 Customer Entity Data Entries

3.2.4 Project

The Project table stores the data of the project with a primary key as Project_ID and has other details such as Description, Date, Budget, and Actual_Cost. Employee_ID and Customer_ID in this table are foreign keys referencing from the Employee and Customer table. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 Project_ID	int(11)			No	None			Change Drop More
☐	2 Description	varchar(255)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	3 Date	date			Yes	NULL			Change Drop More
☐	4 Budget	decimal(10,2)			Yes	NULL			Change Drop More
☐	5 Actual_Cost	decimal(10,2)			Yes	NULL			Change Drop More
☐	6 Employee_ID	int(11)			Yes	NULL			Change Drop More
☐	7 Customer_ID	int(11)			Yes	NULL			Change Drop More

Figure 13 Employee Entity Structure

+ Options

← T →			Project_ID	Description	Date	Budget	Actual_Cost	Employee_ID	Customer_ID
☐ Edit Copy Delete			40001	Website Development	2024-01-10	5000.00	4800.00	20001	30001
☐ Edit Copy Delete			40002	Mobile App Development	2024-02-15	10000.00	9500.00	20002	30002
☐ Edit Copy Delete			40003	Marketing Campaign	2024-03-20	8000.00	7800.00	20003	30003
☐ Edit Copy Delete			40004	Product Launch	2024-04-25	12000.00	11500.00	20004	30004
☐ Edit Copy Delete			40005	Consulting Project	2024-05-30	6000.00	5800.00	20005	30005
☐ Edit Copy Delete			40006	Research Project	2024-06-05	7000.00	6800.00	20006	30006
☐ Edit Copy Delete			40007	Event Planning	2024-07-10	9000.00	8500.00	20007	30007
☐ Edit Copy Delete			40008	Training Program	2024-08-15	3000.00	2800.00	20008	30008
☐ Edit Copy Delete			40009	Software Implementation	2024-09-20	15000.00	14500.00	20009	30009
☐ Edit Copy Delete			40010	E-commerce Integration	2024-10-25	11000.00	10500.00	20010	30010
☐ Edit Copy Delete			40011	Customer Support Improvement	2024-11-30	4000.00	3800.00	20011	30011
☐ Edit Copy Delete			40012	Financial Analysis	2025-01-05	8500.00	8200.00	20012	30012
☐ Edit Copy Delete			40013	Supply Chain Optimization	2025-02-10	9500.00	9200.00	20013	30013
☐ Edit Copy Delete			40014	HR Restructuring	2025-03-15	7500.00	7200.00	20014	30014
☐ Edit Copy Delete			40015	Quality Assurance	2025-04-20	5000.00	4800.00	20015	30015

↑

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Figure 14 Project Entity Data Entries

3.2.5 Task

The Task table stores the data of the task with a primary key as Task_ID and has other details such as Description, Starting_Date, and Ending_date. Plan_ID in this table is a foreign key referencing from the plan table. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 Task_ID	int(11)			No	None			Change Drop More
☐	2 Description	varchar(255)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐	3 Starting_Date	date			Yes	NULL			Change Drop More
☐	4 Ending_Date	date			Yes	NULL			Change Drop More
☐	5 Plan_ID	int(11)			Yes	NULL			Change Drop More

Figure 15 Task Entity Structure

+ Options

← T →				Task_ID	Description	Starting_Date	Ending_Date	Plan_ID
☐	 Edit	 Copy	 Delete	50001	Design UI	2024-01-10	2024-01-20	60001
☐	 Edit	 Copy	 Delete	50002	Develop Frontend	2024-01-20	2024-01-30	60002
☐	 Edit	 Copy	 Delete	50003	Develop Backend	2024-01-30	2024-02-10	60003
☐	 Edit	 Copy	 Delete	50004	Testing	2024-02-10	2024-02-15	60004
☐	 Edit	 Copy	 Delete	50005	Market Research	2024-03-20	2024-03-30	60005
☐	 Edit	 Copy	 Delete	50006	Advertising Campaign	2024-03-30	2024-04-10	60006
☐	 Edit	 Copy	 Delete	50007	Product Launch Event	2024-04-25	2024-04-30	60007
☐	 Edit	 Copy	 Delete	50008	Customer Training	2024-05-30	2024-06-05	60008
☐	 Edit	 Copy	 Delete	50009	Software Development	2024-06-05	2024-06-15	60009
☐	 Edit	 Copy	 Delete	50010	Integration Testing	2024-07-10	2024-07-20	60010
☐	 Edit	 Copy	 Delete	50011	Event Management	2024-08-15	2024-08-25	60011
☐	 Edit	 Copy	 Delete	50012	Training Delivery	2024-09-20	2024-09-30	60012
☐	 Edit	 Copy	 Delete	50013	Implementation Planning	2024-10-25	2024-11-05	60013
☐	 Edit	 Copy	 Delete	50014	Analysis	2024-11-30	2025-01-05	60014
☐	 Edit	 Copy	 Delete	50015	Optimization	2025-02-10	2025-02-20	60015

Figure 16 Task Entity Data Entries

3.2.6 Plan

The Plan table stores the data of the plan with a primary key as Plan_ID and has other details such as Time_Period. Employee_ID in this table is a foreign key referencing from the Employee table. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	Plan_ID	int(11)			No	None			Change Drop More
☐ 2	Time_Period	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐ 3	Employee_ID	int(11)			Yes	NULL			Change Drop More

Figure 17 Plan Entity Structure

+ Options

					Plan_ID	Time_Period	Employee_ID
☐	Edit	Copy	Delete		60001	Weekly	20001
☐	Edit	Copy	Delete		60002	Bi-weekly	20002
☐	Edit	Copy	Delete		60003	Monthly	20003
☐	Edit	Copy	Delete		60004	Weekly	20004
☐	Edit	Copy	Delete		60005	Bi-weekly	20005
☐	Edit	Copy	Delete		60006	Monthly	20006
☐	Edit	Copy	Delete		60007	Weekly	20007
☐	Edit	Copy	Delete		60008	Bi-weekly	20008
☐	Edit	Copy	Delete		60009	Monthly	20009
☐	Edit	Copy	Delete		60010	Weekly	20010
☐	Edit	Copy	Delete		60011	Bi-weekly	20011
☐	Edit	Copy	Delete		60012	Monthly	20012
☐	Edit	Copy	Delete		60013	Weekly	20013
☐	Edit	Copy	Delete		60014	Bi-weekly	20014
☐	Edit	Copy	Delete		60015	Monthly	20015

☐ Check all With selected: Edit Copy De

Figure 18 Plan Entity Data Entries

3.2.7 Skill

The Skill table stores the data of the skill with a primary key as Skill_ID and has other details such as Description, and Rate_Of_Pay. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	Skill_ID	int(11)			No	None			Change Drop More
☐ 2	Description	varchar(255)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐ 3	Rate_of_Pay	decimal(10,2)			Yes	NULL			Change Drop More

Figure 19 Skill Entity Structure

+ Options						
				Skill_ID	Description	Rate_of_Pay
☐	Edit	Copy	Delete	70001	Web Development	50.00
☐	Edit	Copy	Delete	70002	Mobile App Development	60.00
☐	Edit	Copy	Delete	70003	Marketing	45.00
☐	Edit	Copy	Delete	70004	Product Management	55.00
☐	Edit	Copy	Delete	70005	Consulting	70.00
☐	Edit	Copy	Delete	70006	Research	65.00
☐	Edit	Copy	Delete	70007	Event Planning	40.00
☐	Edit	Copy	Delete	70008	Training	30.00
☐	Edit	Copy	Delete	70009	Software Development	75.00
☐	Edit	Copy	Delete	70010	Integration	65.00
☐	Edit	Copy	Delete	70011	Customer Support	35.00
☐	Edit	Copy	Delete	70012	Financial Analysis	80.00
☐	Edit	Copy	Delete	70013	Supply Chain Management	85.00
☐	Edit	Copy	Delete	70014	HR Management	40.00
☐	Edit	Copy	Delete	70015	Quality Assurance	50.00
	☐ Check all	With selected:		Edit	Copy	Delete

Figure 20 Skill Entity Data Entries

3.2.8 Work_Log

The Work_Log table stores the data of the work_log with a primary key as Log_ID and has other details such as Date, and Total_Hours_Worked. Employee_ID and Bill_ID in this table are foreign keys referencing from the Employee and Bill table. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	Log_ID	int(11)			No	None			Change Drop More
☐ 2	Date	date			Yes	NULL			Change Drop More
☐ 3	Total_Hours_Worked	decimal(6,2)			Yes	NULL			Change Drop More
☐ 4	Employee_ID	int(11)			Yes	NULL			Change Drop More
☐ 5	Bill_ID	int(11)			Yes	NULL			Change Drop More

Figure 21 Worklog Entity Data Structure

+ Options

← T →				Log_ID	Date	Total_Hours_Worked	Employee_ID	Bill_ID			
☐		Edit		Copy		Delete	80001	2024-01-10	8.00	20001	90001
☐		Edit		Copy		Delete	80002	2024-01-20	8.00	20002	90002
☐		Edit		Copy		Delete	80003	2024-01-30	8.00	20003	90003
☐		Edit		Copy		Delete	80004	2024-02-10	8.00	20004	90004
☐		Edit		Copy		Delete	80005	2024-03-20	8.00	20005	90005
☐		Edit		Copy		Delete	80006	2024-03-30	8.00	20006	90006
☐		Edit		Copy		Delete	80007	2024-04-25	8.00	20007	90007
☐		Edit		Copy		Delete	80008	2024-05-30	8.00	20008	90008
☐		Edit		Copy		Delete	80009	2024-06-05	8.00	20009	90009
☐		Edit		Copy		Delete	80010	2024-07-10	8.00	20010	90010
☐		Edit		Copy		Delete	80011	2024-08-15	8.00	20011	90011
☐		Edit		Copy		Delete	80012	2024-09-20	8.00	20012	90012
☐		Edit		Copy		Delete	80013	2024-10-25	8.00	20013	90013
☐		Edit		Copy		Delete	80014	2024-11-30	8.00	20014	90014
☐		Edit		Copy		Delete	80015	2025-02-10	8.00	20015	90015

⬆

☐ Check all

With selected:

 Edit

 Copy

 Delete

 Export

Figure 22 Worklog Entity Data Entries

3.2.9 Bill

The Bill table stores the data of the bill with a primary key as Bill_ID and has other details such as Billing_Period, and Total_Hours_Worked. Customer_ID in this table is a foreign key referencing from the Customer table. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	Bill_ID	int(11)			No	None			Change Drop More
☐ 2	Billing_Period	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More
☐ 3	Total_Hours_Worked	decimal(6,2)			Yes	NULL			Change Drop More
☐ 4	Customer_ID	int(11)			Yes	NULL			Change Drop More

Figure 23 Bill Entity Structure

+ Options

  				Bill_ID	Billing_Period	Total_Hours_Worked	Customer_ID
☐	 Edit	 Copy	 Delete	90001	January 2024	40.00	30001
☐	 Edit	 Copy	 Delete	90002	February 2024	40.00	30002
☐	 Edit	 Copy	 Delete	90003	March 2024	40.00	30003
☐	 Edit	 Copy	 Delete	90004	April 2024	40.00	30004
☐	 Edit	 Copy	 Delete	90005	May 2024	40.00	30005
☐	 Edit	 Copy	 Delete	90006	June 2024	40.00	30006
☐	 Edit	 Copy	 Delete	90007	July 2024	40.00	30007
☐	 Edit	 Copy	 Delete	90008	August 2024	40.00	30008
☐	 Edit	 Copy	 Delete	90009	September 2024	40.00	30009
☐	 Edit	 Copy	 Delete	90010	October 2024	40.00	30010
☐	 Edit	 Copy	 Delete	90011	November 2024	40.00	30011
☐	 Edit	 Copy	 Delete	90012	December 2024	40.00	30012
☐	 Edit	 Copy	 Delete	90013	January 2025	40.00	30013
☐	 Edit	 Copy	 Delete	90014	February 2025	40.00	30014
☐	 Edit	 Copy	 Delete	90015	April 2025	40.00	30015


☐ Check all
 With selected:
  Edit
  Copy
  Delete
  Export

Figure 24 Bill Entity Data Entries

3.2.10 Employee_Project (works_on)

The Employee_Project table has the M: N relationship between employee and project (many employees can work on the same project and vice-versa). Employee_ID and Project_ID forms a primary key named Primary and Project_ID as a foreign key. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 Employee_ID	int(11)			No	None			Change Drop More
☐	2 Project_ID	int(11)			No	None			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Ind

Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after Project_ID Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	Employee_ID	15	A	No	
					Project_ID	15	A	No	
Edit Rename Drop	Project_ID	BTREE	No	No	Project_ID	15	A	No	

Figure 25 Employee_Project Relation Structure

+ Options

				Employee_ID	Project_ID
☐	Edit	Copy	Delete	20001	40001
☐	Edit	Copy	Delete	20002	40002
☐	Edit	Copy	Delete	20003	40003
☐	Edit	Copy	Delete	20004	40004
☐	Edit	Copy	Delete	20005	40005
☐	Edit	Copy	Delete	20006	40006
☐	Edit	Copy	Delete	20007	40007
☐	Edit	Copy	Delete	20008	40008
☐	Edit	Copy	Delete	20009	40009
☐	Edit	Copy	Delete	20010	40010
☐	Edit	Copy	Delete	20011	40011
☐	Edit	Copy	Delete	20012	40012
☐	Edit	Copy	Delete	20013	40013
☐	Edit	Copy	Delete	20014	40014
☐	Edit	Copy	Delete	20015	40015

☐ Check all With selected: Edit Copy

Figure 26 Employee_Project Relation Data Entries

3.2.11 Employee_Task (assigned)

The Employee_Task table has the M: N relationship between employee and Task (many employees can be assigned the same task and vice-versa). Employee_ID and Task_ID forms a primary key named Primary and Task_ID as a foreign key. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐ 1	Employee_ID	int(11)			No	None			Change Drop More
☐ 2	Task_ID	int(11)			No	None			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index
 Remove from central columns

Print Propose table structure Track table Move columns Normalize
 Add 1 column(s) after Task_ID Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	Employee_ID Task_ID	15 15	A A	No No	
Edit Rename Drop	Task_ID	BTREE	No	No	Task_ID	15	A	No	

Figure 27 Employee_Task Relation Structure

+ Options
 Employee_ID Task_ID

☐ Edit Copy Delete	20001	50001
☐ Edit Copy Delete	20002	50002
☐ Edit Copy Delete	20003	50003
☐ Edit Copy Delete	20004	50004
☐ Edit Copy Delete	20005	50005
☐ Edit Copy Delete	20006	50006
☐ Edit Copy Delete	20007	50007
☐ Edit Copy Delete	20008	50008
☐ Edit Copy Delete	20009	50009
☐ Edit Copy Delete	20010	50010
☐ Edit Copy Delete	20011	50011
☐ Edit Copy Delete	20012	50012
☐ Edit Copy Delete	20013	50013
☐ Edit Copy Delete	20014	50014
☐ Edit Copy Delete	20015	50015

☐ Check all With selected: Edit Co

Figure 28 Employee_Task Relation Data Entries

3.2.12 Employee_Skill (has)

The Employee_Skill table has the M: N relationship between employee and skill (one employee has many skills and vice-versa). Employee_ID and Skill_ID forms a primary key named Primary and Skill_ID as a foreign key. Below are the figures showing the structure and stored data:

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1 Employee_ID	int(11)			No	None			Change Drop More
☐	2 Skill_ID	int(11)			No	None			Change Drop More

☐ Check all With selected: Browse Change Drop Primary Unique Index
 Remove from central columns

Print Propose table structure Track table Move columns Normalize

Add 1 column(s) after Skill_ID Go

Indexes

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	PRIMARY	BTREE	Yes	No	Employee_ID Skill_ID	15	A	No	No
Edit Rename Drop	Skill_ID	BTREE	No	No	Skill_ID	15	A	No	No

Figure 29 Employee_Skill Relation Structure

+ Options

				Employee_ID	Skill_ID
☐	Edit	Copy	Delete	20001	70001
☐	Edit	Copy	Delete	20002	70002
☐	Edit	Copy	Delete	20003	70003
☐	Edit	Copy	Delete	20004	70004
☐	Edit	Copy	Delete	20005	70005
☐	Edit	Copy	Delete	20006	70006
☐	Edit	Copy	Delete	20007	70007
☐	Edit	Copy	Delete	20008	70008
☐	Edit	Copy	Delete	20009	70009
☐	Edit	Copy	Delete	20010	70010
☐	Edit	Copy	Delete	20011	70011
☐	Edit	Copy	Delete	20012	70012
☐	Edit	Copy	Delete	20013	70013
☐	Edit	Copy	Delete	20014	70014
☐	Edit	Copy	Delete	20015	70015

☐ Check all With selected: Edit Copy

Figure 30 Employee_Skill Relation Data Entries

3.2.13 Plan_Type

The Plan_Type table is formed by the multivalued attribute relation with the Plan entity and has Plan_ID as a foreign key with no primary key due to multiple value entries on the same Plan_ID. This also shows the normalization technique used in this case. Below are the figures showing the structure and stored data:

The screenshot displays a database management tool interface. At the top, the table structure for 'Plan_Type' is shown with two columns: 'Plan_ID' (int(11)) and 'Type' (varchar(50)). Below this, there are options to 'Check all', 'With selected:', 'Browse', 'Change', 'Drop', 'Primary', 'Unique', and 'Index'. A 'Remove from central columns' button is also present. Below the table structure, there are options to 'Print', 'Propose table structure', 'Track table', 'Move columns', and 'Normalize'. A search bar shows '1' column(s) 'after Type' with a 'Go' button. Below this, the 'Indexes' tab is selected, showing a table with columns: Action, Keyname, Type, Unique, Packed, Column, Cardinality, Collation, Null, and Comment. The table lists one index for 'Plan_ID' with a BTREE type, unique, not packed, cardinality of 20, collation 'A', and null 'Yes'.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
1	Plan_ID	int(11)			Yes	NULL			Change Drop More
2	Type	varchar(50)	latin1_swedish_ci		Yes	NULL			Change Drop More

Action	Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
Edit Rename Drop	Plan_ID	BTREE	No	No	Plan_ID	20	A	Yes	

Figure 31 Plan_Type Data Structure

[+ Options](#)

Plan_ID	Type
60001	Design
60002	Development
60003	Design
60003	Development
60004	Development
60005	Design
60006	Development
60007	Design
60007	Development
60008	Design
60009	Development
60010	Design
60010	Development
60011	Development
60012	Design
60013	Development
60013	Design
60014	Design
60014	Development
60015	Design

Figure 32 Plan_Type Data Entries

3.3 Queries Examples

MySQL Queries

- 1) Find all the stakeholders belonging to the region of North America.

```
SELECT * FROM stakeholder WHERE Region = 'North America';
```

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050 » Table: stakeholder

Browser Structure SQL Search Insert Export Import Privileges Operations

Show query box

Showing rows 0 - 5 (6 total, Query took 0.0009 seconds.)

```
SELECT * FROM stakeholder WHERE Region = 'North America';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

				Stakeholder_ID	First_Name	Middle_Initial	Last_Name	Region	Phone_Number
☐	Edit	Copy	Delete	10001	Robert	A	Wilson	North America	555-1234
☐	Edit	Copy	Delete	10008	Sarah	H	White	North America	555-0123
☐	Edit	Copy	Delete	10014	Michelle	N	Young	North America	555-5678
☐	Edit	Copy	Delete	10016	Maria	A	Gonzalez	North America	555-1234
☐	Edit	Copy	Delete	10023	Juan	H	Vargas	North America	555-0123
☐	Edit	Copy	Delete	10029	Gloria	N	Moreno	North America	555-5678

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

- 2) Find all the employees hired after January 1, 2024.

```
SELECT * FROM Employee WHERE Date_of_Hire > '2024-01-01';
```

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050 » Table: Employee

Browser Structure SQL Search Insert Export Import Privileges Operations Tracking Tri

Show query box

Showing rows 0 - 4 (5 total, Query took 0.0011 seconds.)

```
SELECT * FROM Employee WHERE Date_of_Hire > '2024-01-01';
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

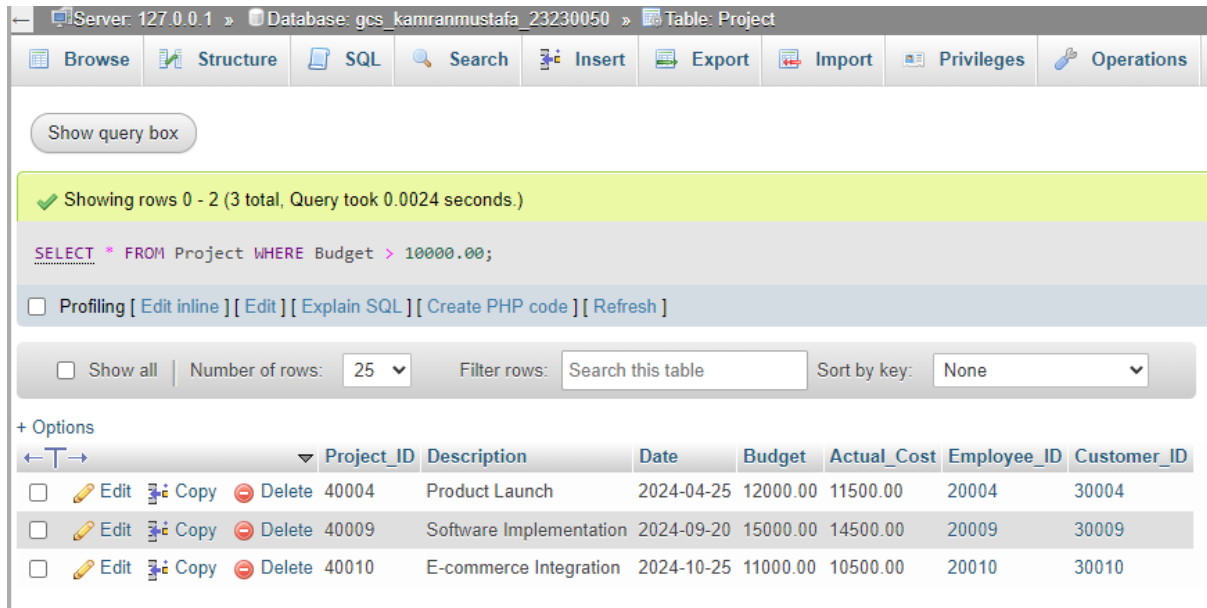
Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

				Employee_ID	First_Name	Middle_Initial	Last_Name	Region	Date_of_Hire	Phone_Number	Stakeholder_ID
☐	Edit	Copy	Delete	20011	Michael	K	Roberts	Africa	2024-01-05	555-2345	10011
☐	Edit	Copy	Delete	20012	Emily	L	Thomas	South America	2024-02-10	555-6789	10012
☐	Edit	Copy	Delete	20013	Christopher	M	Hill	Europe	2024-03-20	555-1234	10013
☐	Edit	Copy	Delete	20014	Michelle	N	Young	North America	2024-04-15	555-5678	10014
☐	Edit	Copy	Delete	20015	Joseph	O	King	Asia	2024-05-05	555-9012	10015

3) Find all the projects with a budget greater than \$10,000.

```
SELECT * FROM Project WHERE Budget > 10000.00;
```



Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050 » Table: Project

Showing rows 0 - 2 (3 total, Query took 0.0024 seconds.)

`SELECT * FROM Project WHERE Budget > 10000.00;`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

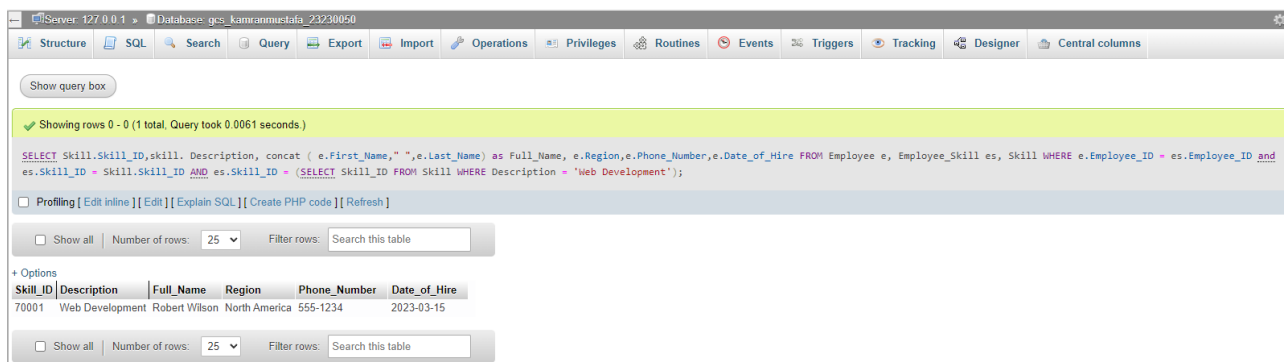
☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

+ Options

	Project_ID	Description	Date	Budget	Actual_Cost	Employee_ID	Customer_ID
☐ Edit Copy Delete	40004	Product Launch	2024-04-25	12000.00	11500.00	20004	30004
☐ Edit Copy Delete	40009	Software Implementation	2024-09-20	15000.00	14500.00	20009	30009
☐ Edit Copy Delete	40010	E-commerce Integration	2024-10-25	11000.00	10500.00	20010	30010

4) Find all the employees who have a specific skill like “Web Development”.

```
SELECT Skill.Skill_ID, skill. Description, concat ( e.First_Name, " ", e.Last_Name) as Full_Name, e.Region, e.Phone_Number, e.Date_of_Hire
FROM Employee e, Employee_Skill es, Skill
WHERE e.Employee_ID = es.Employee_ID
and es.Skill_ID = Skill.Skill_ID
AND es.Skill_ID = (SELECT Skill_ID FROM Skill WHERE Description = 'Web Development');
```



Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050

Showing rows 0 - 0 (1 total, Query took 0.0061 seconds.)

`SELECT Skill.Skill_ID, skill. Description, concat (e.First_Name, " ", e.Last_Name) as Full_Name, e.Region, e.Phone_Number, e.Date_of_Hire FROM Employee e, Employee_Skill es, Skill WHERE e.Employee_ID = es.Employee_ID and es.Skill_ID = Skill.Skill_ID AND es.Skill_ID = (SELECT Skill_ID FROM Skill WHERE Description = 'Web Development');`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

Skill_ID	Description	Full_Name	Region	Phone_Number	Date_of_Hire
70001	Web Development	Robert Wilson	North America	555-1234	2023-03-15

☐ Show all | Number of rows: 25 | Filter rows: Search this table

5) Find the customers who have spent a good budget on the projects.

```
SELECT c.Customer_ID, concat(c.First_Name, " ", c.Last_Name) as Full_Name,
SUM(p.Actual_Cost) AS Total_Spending
FROM Customer c, Project p
WHERE c.Customer_ID = p.Customer_ID
GROUP BY c.Customer_ID
ORDER BY Total_Spending DESC;
```

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050 » Table: Customer

Showing rows 0 - 14 (15 total, Query took 0.0028 seconds)

```
SELECT c.Customer_ID, concat(c.First_Name, " ", c.Last_Name) as Full_Name, SUM(p.Actual_Cost) AS Total_Spending FROM Customer c, Project p WHERE c.Customer_ID = p.Customer_ID GROUP BY c.Customer_ID ORDER BY Total_Spending DESC;
```

Options: Customer_ID, Full_Name, Total_Spending

Customer_ID	Full_Name	Total_Spending
30009	Luis Lopez	14500.00
30004	Carlos Rivera	11500.00
30010	Miguel Martinez	10500.00
30002	David Perez	9500.00
30013	Javier Fernandez	9200.00
30007	Ana Torres	8500.00
30012	Rosa Hernandez	8200.00
30003	Mary Sanchez	7800.00
30014	Gloria Moreno	7200.00
30006	Pedro Diaz	6800.00
30005	Laura Gomez	5800.00
30001	Maria Gonzalez	4800.00
30015	Sofia Alvarez	4800.00
30011	Elena Garcia	3800.00
30008	Juan Vargas	2800.00

6) Retrieve all projects that are behind schedule (the end date has passed).

```
SELECT *
FROM Project
WHERE Date < CURDATE();
```

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050 » Table: Project

Showing rows 0 - 3 (4 total, Query took 0.0009 seconds)

```
SELECT * FROM Project WHERE Date < CURDATE();
```

Options: Project_ID, Description, Date, Budget, Actual_Cost, Employee_ID, Customer_ID

Project_ID	Description	Date	Budget	Actual_Cost	Employee_ID	Customer_ID
40001	Website Development	2024-01-10	5000.00	4800.00	20001	30001
40002	Mobile App Development	2024-02-15	10000.00	9500.00	20002	30002
40003	Marketing Campaign	2024-03-20	8000.00	7800.00	20003	30003
40004	Product Launch	2024-04-25	12000.00	11500.00	20004	30004

7) Check if there is a project with the highest budget among all the projects.

```
SELECT *
FROM Project
WHERE Budget = (SELECT MAX(Budget) FROM Project);
```

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050 » Table: Project

Browse Structure SQL Search Insert Export Import Privileges Operations

Show query box

Showing rows 0 - 0 (1 total, Query took 0.0021 seconds.)

```
SELECT * FROM Project WHERE Budget = (SELECT MAX(Budget) FROM Project);
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

	Project_ID	Description	Date	Budget	Actual_Cost	Employee_ID	Customer_ID
☐ Edit ☐ Copy ☐ Delete	40009	Software Implementation	2024-09-20	15000.00	14500.00	20009	30009

Check all | With selected: Edit Copy Delete Export

8) Please check the tasks assigned to each employee to complete a project.

```
SELECT e.First_Name, e.Last_Name, t.Description AS Task_Description
FROM employee e, employee_task et, task t
WHERE e.Employee_ID = et.Employee_ID
AND et.Task_ID = t.Task_ID;
```

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking Designer

Show query box

Showing rows 0 - 14 (15 total, Query took 0.0021 seconds.)

```
SELECT e.First_Name, e.Last_Name, t.Description AS Task_Description FROM employee e, employee_task et, task t WHERE e.Employee_ID = et.Employee_ID AND et.Task_ID = t.Task_ID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

First_Name	Last_Name	Task_Description
Robert	Wilson	Design UI
Jennifer	Anderson	Develop Frontend
Christopher	Thompson	Develop Backend
Jessica	Harris	Testing
Matthew	Taylor	Market Research
Amanda	Moore	Advertising Campaign
David	Jackson	Product Launch Event
Sarah	White	Customer Training
Daniel	Adams	Software Development
Ashley	Nelson	Integration Testing
Michael	Roberts	Event Management
Emily	Thomas	Training Delivery
Christopher	Hill	Implementation Planning
Michelle	Young	Analysis
Joseph	King	Optimization

9) Find the plan type suitable for each project to be completed.

```
SELECT p.Description AS Project_Description, pt.Type AS Plan_Type
FROM project p, plan_type pt, plan pl
WHERE p.Employee_ID = pl.Employee_ID
AND pl.Plan_ID = pt.Plan_ID;
```

Server: 127.0.0.1 » Database: ges_kamranmustafa_23230050

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking Design

Show query box

Showing rows 0 - 24 (40 total. Query took 0.0025 seconds.)

```
SELECT p.Description AS Project_Description, pt.Type AS Plan_Type FROM project p, plan_type pt, plan pl WHERE p.Employee_ID = pl.Employee_ID AND pl.Plan_ID = pt.Plan_ID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all Number of rows: 25 Filter rows: Search this table

+ Options

Project_Description	Plan_Type
Website Development	Design
Website Development	Design
Mobile App Development	Development
Mobile App Development	Development
Marketing Campaign	Design
Marketing Campaign	Development
Marketing Campaign	Design
Marketing Campaign	Development
Product Launch	Development
Product Launch	Development
Consulting Project	Design
Consulting Project	Design
Research Project	Development
Research Project	Development
Event Planning	Design
Event Planning	Development
Event Planning	Design
Event Planning	Development
Training Program	Design
Training Program	Design
Software Implementation	Development
Software Implementation	Development
E-commerce Integration	Design
E-commerce Integration	Development
E-commerce Integration	Design

10) Find the customers for whom each employee has worked on the projects.

```
SELECT concat(e.First_Name," ",e.Last_Name) as Employee_Name,
concat(c.First_Name," ",c.Last_Name) as Customer_Name, p.Description
FROM employee e, project p, customer c
WHERE e.Employee_ID = p.Employee_ID
AND p.Customer_ID = c.Customer_ID;
```

Server: 127.0.0.1 » Database: ges_kamranmustafa_23230050 » Table: project

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Show query box

Showing rows 0 - 14 (15 total. Query took 0.0027 seconds.)

```
SELECT concat(e.First_Name," ",e.Last_Name) as Employee_Name, concat(c.First_Name," ",c.Last_Name) as Customer_Name, p.Description FROM employee e, project p, customer c WHERE e.Employee_ID = p.Employee_ID AND p.Customer_ID = c.Customer_ID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all Number of rows: 25 Filter rows: Search this table

+ Options

	Employee_Name	Customer_Name	Description
☐ Edit Copy Delete	Robert Wilson	Maria Gonzalez	Website Development
☐ Edit Copy Delete	Jennifer Anderson	David Perez	Mobile App Development
☐ Edit Copy Delete	Christopher Thompson	Mary Sanchez	Marketing Campaign
☐ Edit Copy Delete	Jessica Harris	Carlos Rivera	Product Launch
☐ Edit Copy Delete	Matthew Taylor	Laura Gomez	Consulting Project
☐ Edit Copy Delete	Amanda Moore	Pedro Diaz	Research Project
☐ Edit Copy Delete	David Jackson	Ana Torres	Event Planning
☐ Edit Copy Delete	Sarah White	Juan Vargas	Training Program
☐ Edit Copy Delete	Daniel Adams	Luis Lopez	Software Implementation
☐ Edit Copy Delete	Ashley Nelson	Miguel Martinez	E-commerce Integration
☐ Edit Copy Delete	Michael Roberts	Elena Garcia	Customer Support Improvement
☐ Edit Copy Delete	Emily Thomas	Rosa Hernandez	Financial Analysis
☐ Edit Copy Delete	Christopher Hill	Javier Fernandez	Supply Chain Optimization
☐ Edit Copy Delete	Michelle Young	Gloria Moreno	HR Restructuring
☐ Edit Copy Delete	Joseph King	Sofia Alvarez	Quality Assurance

Check all With selected: Edit Copy Delete Export

11) Find the skills possessed by each employee.

```
SELECT e.First_Name, e.Last_Name, s.Description AS Skill_Description
FROM employee e, employee_skill es, skill s
WHERE e.Employee_ID = es.Employee_ID
```

AND es.Skill_ID = s.Skill_ID;

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Tracking Designer

Show query box

Showing rows 0 - 14 (15 total. Query took 0.0023 seconds.)

```
SELECT e.First_Name, e.Last_Name, s.Description AS Skill_Description FROM employee e, employee_skill es, skill s WHERE e.Employee_ID = es.Employee_ID AND es.Skill_ID = s.Skill_ID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

First_Name	Last_Name	Skill_Description
Robert	Wilson	Web Development
Jennifer	Anderson	Mobile App Development
Christopher	Thompson	Marketing
Jessica	Harris	Product Management
Matthew	Taylor	Consulting
Amanda	Moore	Research
David	Jackson	Event Planning
Sarah	White	Training
Daniel	Adams	Software Development
Ashley	Nelson	Integration
Michael	Roberts	Customer Support
Emily	Thomas	Financial Analysis
Christopher	Hill	Supply Chain Management
Michelle	Young	HR Management
Joseph	King	Quality Assurance

12) Find the total hours worked by each employee.

```
SELECT e.First_Name, e.Last_Name, SUM(wl.Total_Hours_Worked) AS
Total_Hours_Worked
FROM employee e, work_log wl
WHERE e.Employee_ID = wl.Employee_ID
GROUP BY e.Employee_ID;
```

Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050 » Table: employee

Browse Structure SQL Search Insert Export Import Privileges Operations Tracking Triggers

Show query box

Showing rows 0 - 14 (15 total. Query took 0.0020 seconds.)

```
SELECT e.First_Name, e.Last_Name, SUM(wl.Total_Hours_Worked) AS Total_Hours_Worked FROM employee e, work_log wl WHERE e.Employee_ID = wl.Employee_ID GROUP BY e.Employee_ID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

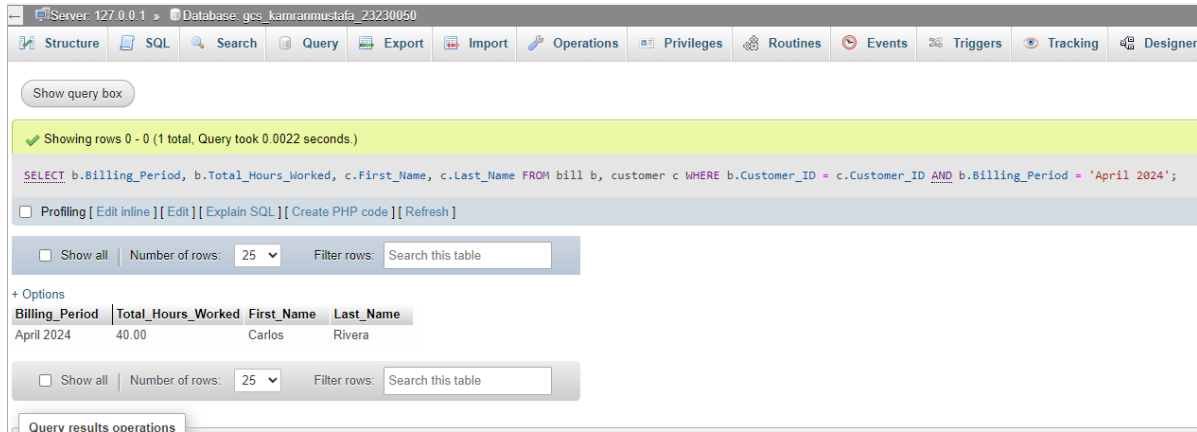
Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

	First_Name	Last_Name	Total_Hours_Worked
☐ Edit Copy Delete	Robert	Wilson	8.00
☐ Edit Copy Delete	Jennifer	Anderson	8.00
☐ Edit Copy Delete	Christopher	Thompson	8.00
☐ Edit Copy Delete	Jessica	Harris	8.00
☐ Edit Copy Delete	Matthew	Taylor	8.00
☐ Edit Copy Delete	Amanda	Moore	8.00
☐ Edit Copy Delete	David	Jackson	8.00
☐ Edit Copy Delete	Sarah	White	8.00
☐ Edit Copy Delete	Daniel	Adams	8.00
☐ Edit Copy Delete	Ashley	Nelson	8.00
☐ Edit Copy Delete	Michael	Roberts	8.00
☐ Edit Copy Delete	Emily	Thomas	8.00
☐ Edit Copy Delete	Christopher	Hill	8.00
☐ Edit Copy Delete	Michelle	Young	8.00
☐ Edit Copy Delete	Joseph	King	8.00

13) Retrieve the billing information for projects completed in April 2024.

```
SELECT b.Billing_Period, b.Total_Hours_Worked, c.First_Name, c.Last_Name
FROM bill b, customer c
WHERE b.Customer_ID = c.Customer_ID
AND b.Billing_Period = 'April 2024';
```



4. Security Scenario

Any database must have data security since it safeguards the availability, confidentiality, and integrity of its users. Security precautions in MySQL databases are essential for protecting private data and guaranteeing that only authorized users can access the database. This entails putting strong authentication measures in place, like passwords and usernames, and using encryption for both data storage and transmission. Furthermore, MySQL offers tools such as role-based access control (RBAC) and access control lists (ACLs) to efficiently manage permissions, restricting access to data according to user roles and privileges. (DuBois, P., 2013)

There are several steps to achieve a security scenario: Create roles > Grant privileges to the created role > Create users > Grant role to the users.

4.1 Create roles and grant privileges

Four different roles are created for the Global Computer Solutions (GCS) database security as named 'gcs_admin', 'gcs_manager', 'gcs_customer', 'gcs_customer'.

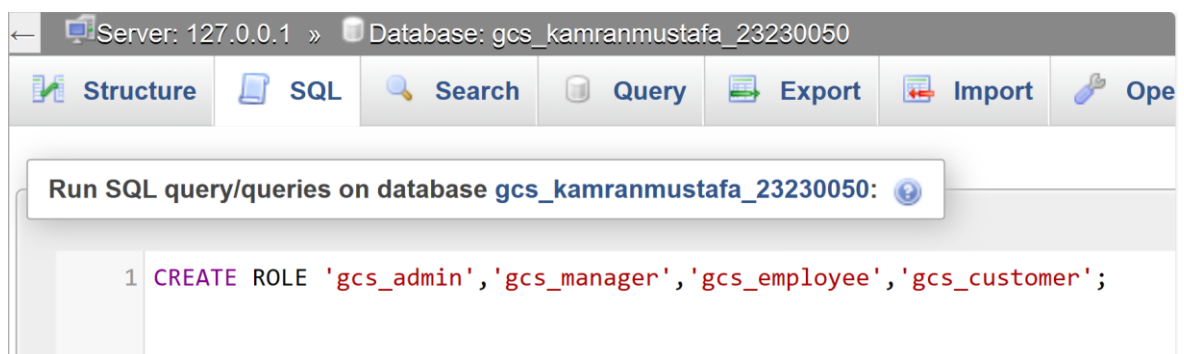
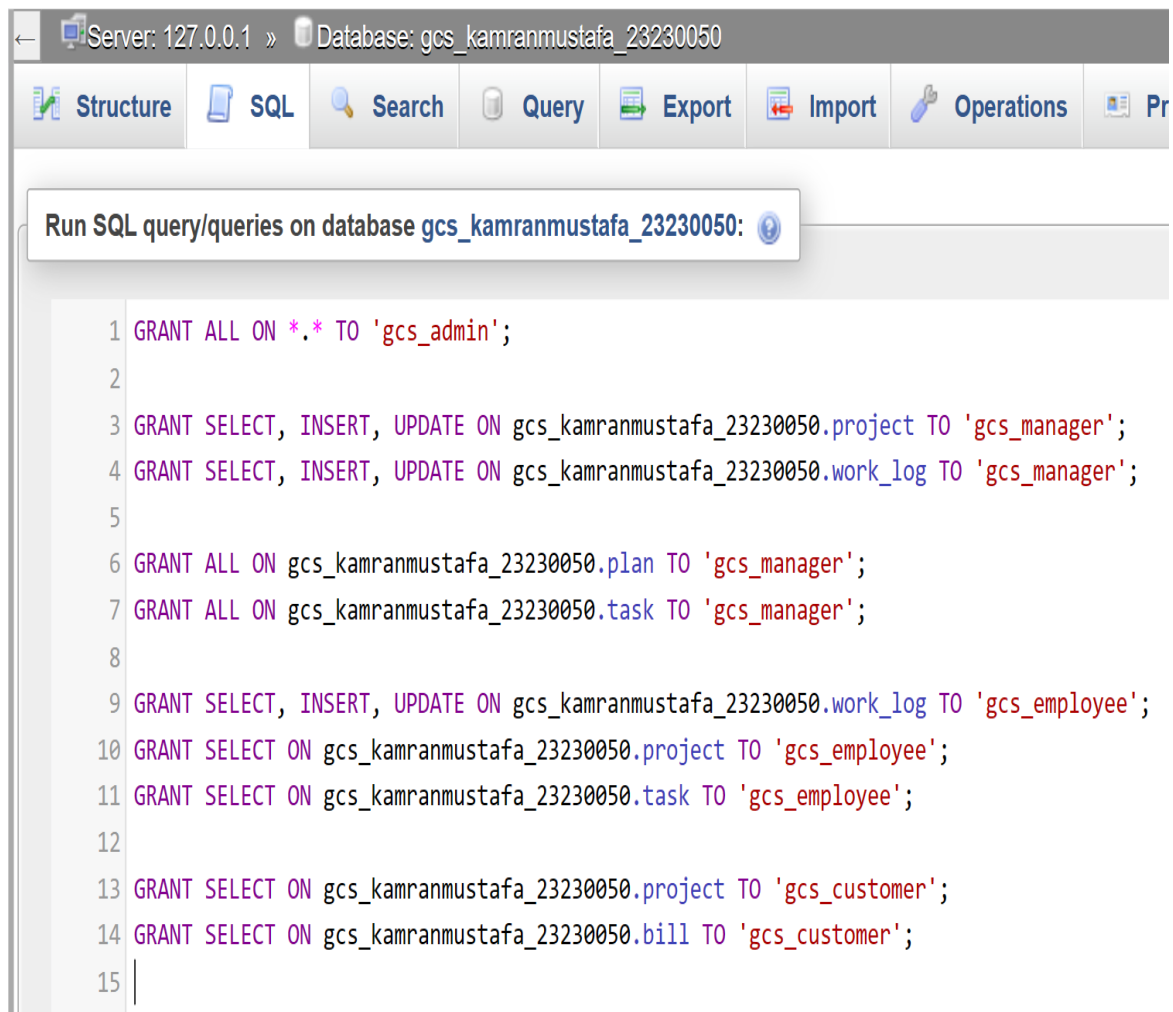


Figure 33 Create Roles

Privileges Given to role:

1. All privileges are given to 'gcs_admin' so, the admin can do anything with the database named 'gcs_kamranmustafa_23230050'.
2. The 'gcs_manager' is responsible for managing projects as well as maintaining the work_log so, granted SELECT, INSERT, and update on gcs_kamranmustafa_23230050 for work_log and project tables.
3. All privileges are given to 'gcs_manager' on the plan and task table because a manager can assign tasks make plans and perform any operations with them.
4. The 'gcs_employee' can only view project details and task details. For the work_log table, they can INSERT and UPDATE in it.
5. The 'gcs_customer' only has viewer rights to view bill and project details.



The screenshot shows a database management tool interface. At the top, there's a header bar with a back arrow, a server icon, and the text 'Server: 127.0.0.1 » Database: gcs_kamranmustafa_23230050'. Below this is a toolbar with icons and labels for 'Structure', 'SQL', 'Search', 'Query', 'Export', 'Import', 'Operations', and 'Pr'. A central panel has a title bar that says 'Run SQL query/queries on database gcs_kamranmustafa_23230050:'. Below the title bar is a text area containing 15 lines of SQL code, with line numbers 1 through 15 on the left. The code consists of several GRANT statements for different users and tables.

```
1 GRANT ALL ON *.* TO 'gcs_admin';
2
3 GRANT SELECT, INSERT, UPDATE ON gcs_kamranmustafa_23230050.project TO 'gcs_manager';
4 GRANT SELECT, INSERT, UPDATE ON gcs_kamranmustafa_23230050.work_log TO 'gcs_manager';
5
6 GRANT ALL ON gcs_kamranmustafa_23230050.plan TO 'gcs_manager';
7 GRANT ALL ON gcs_kamranmustafa_23230050.task TO 'gcs_manager';
8
9 GRANT SELECT, INSERT, UPDATE ON gcs_kamranmustafa_23230050.work_log TO 'gcs_employee';
10 GRANT SELECT ON gcs_kamranmustafa_23230050.project TO 'gcs_employee';
11 GRANT SELECT ON gcs_kamranmustafa_23230050.task TO 'gcs_employee';
12
13 GRANT SELECT ON gcs_kamranmustafa_23230050.project TO 'gcs_customer';
14 GRANT SELECT ON gcs_kamranmustafa_23230050.bill TO 'gcs_customer';
15
```

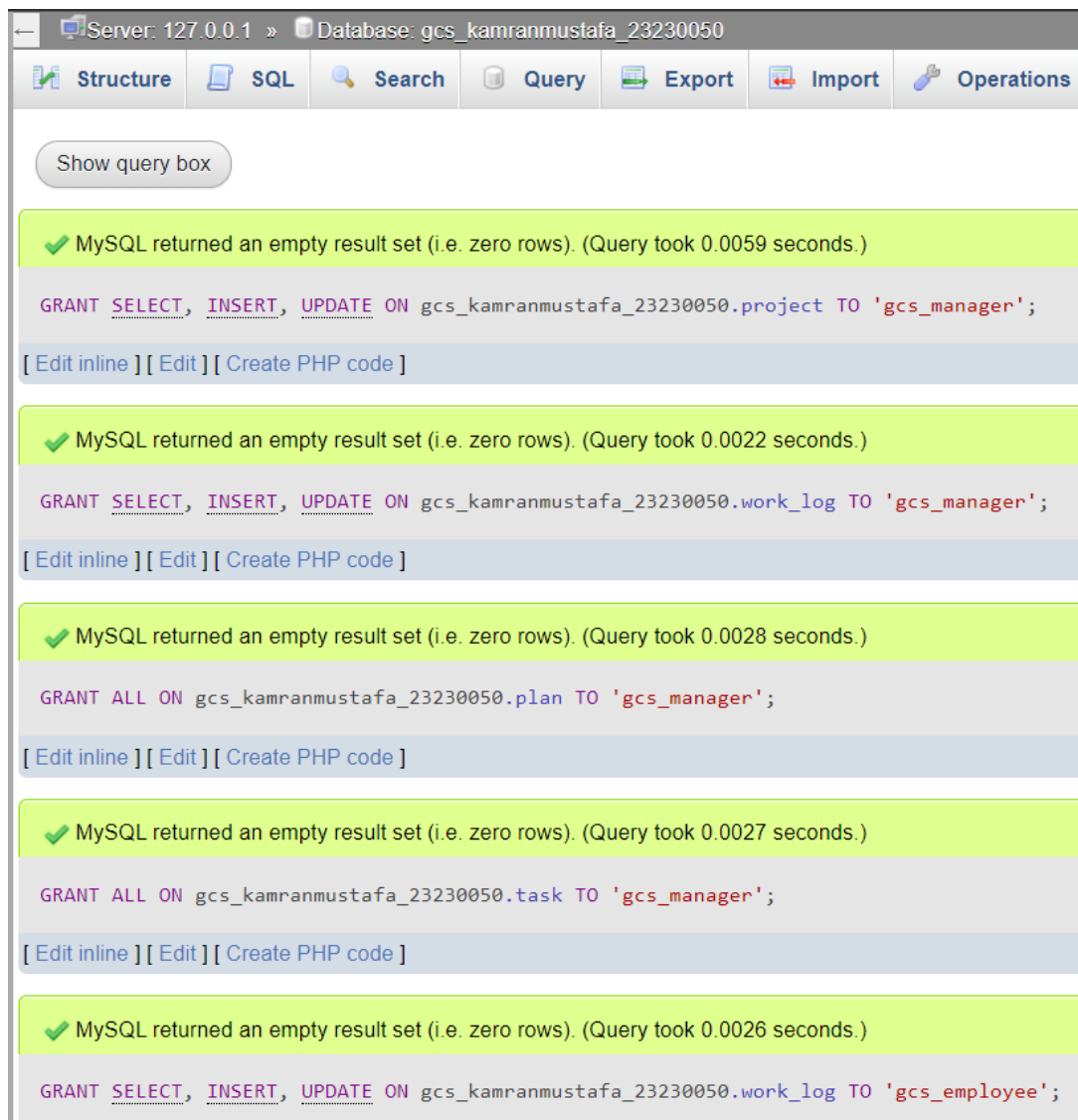



Figure 34 Grant Privileges to the Roles

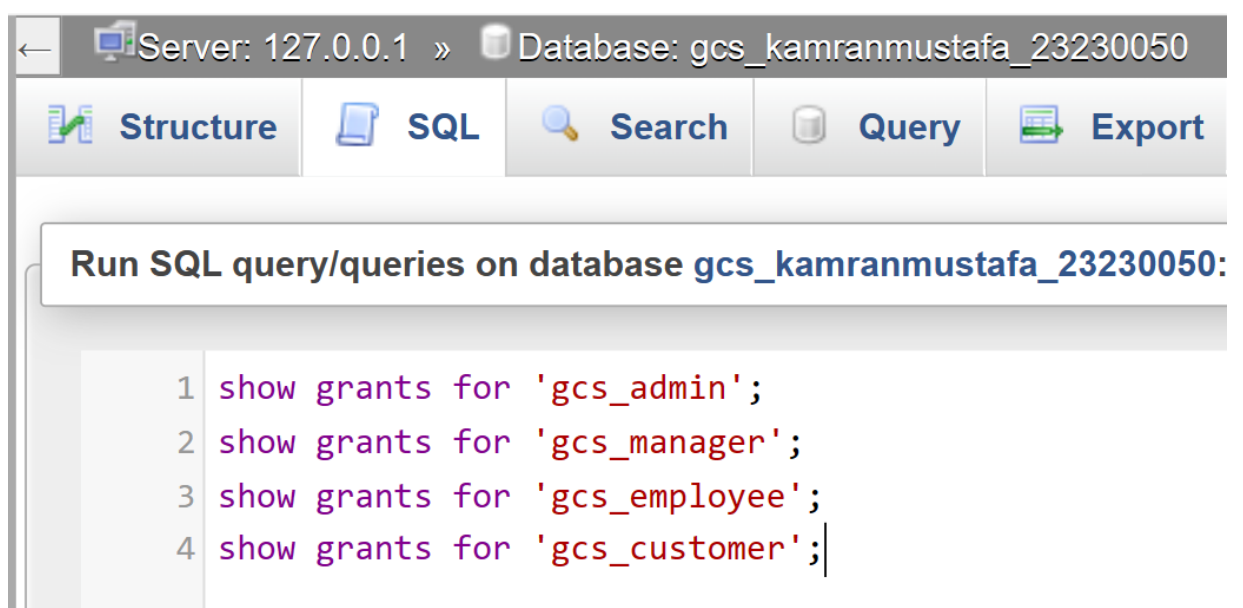


Figure 35 Show Grants for the Roles

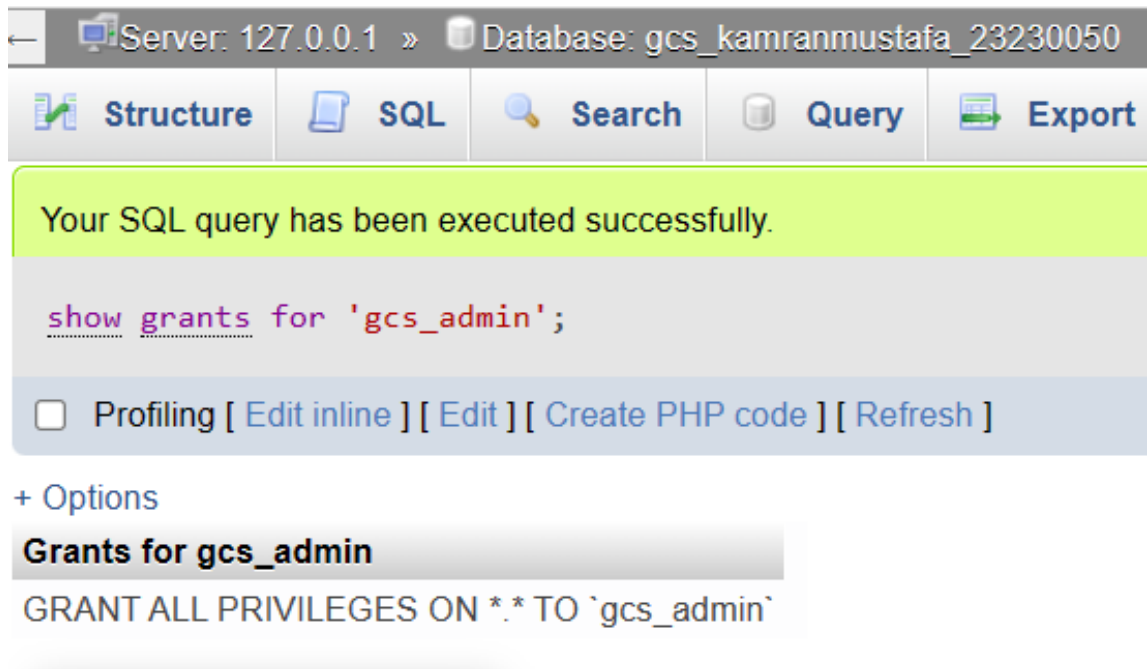


Figure 36 Grant for gcs_admin role

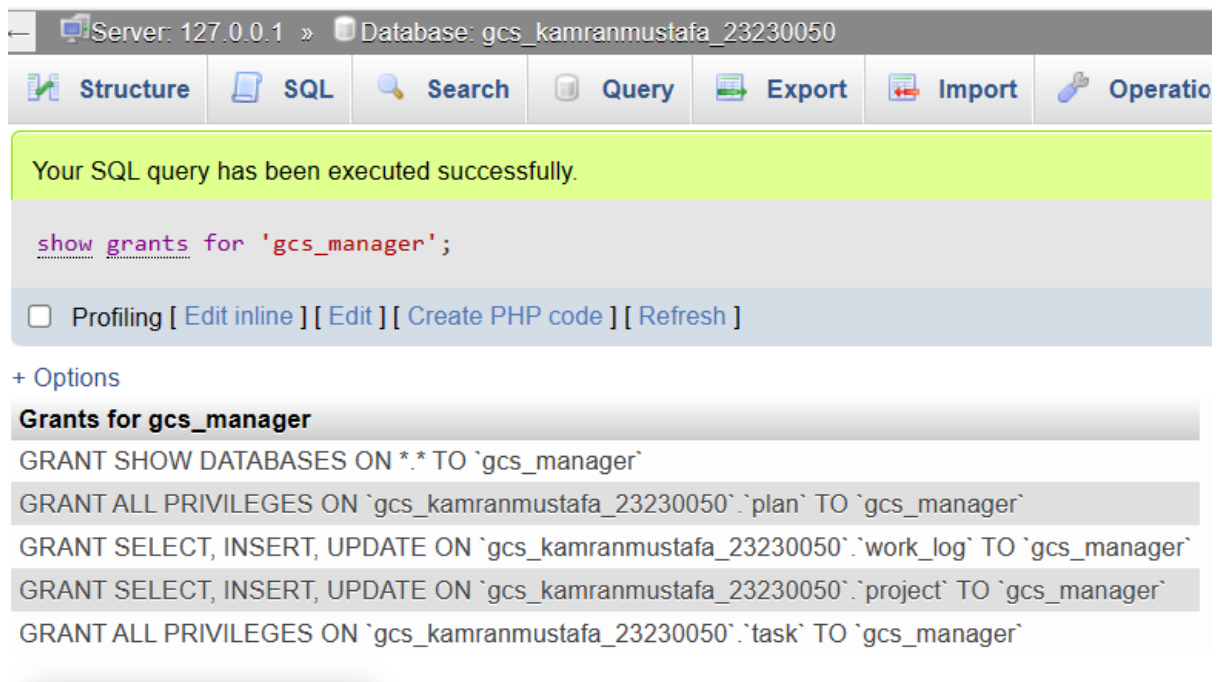


Figure 37 Grant for gcs_manager role

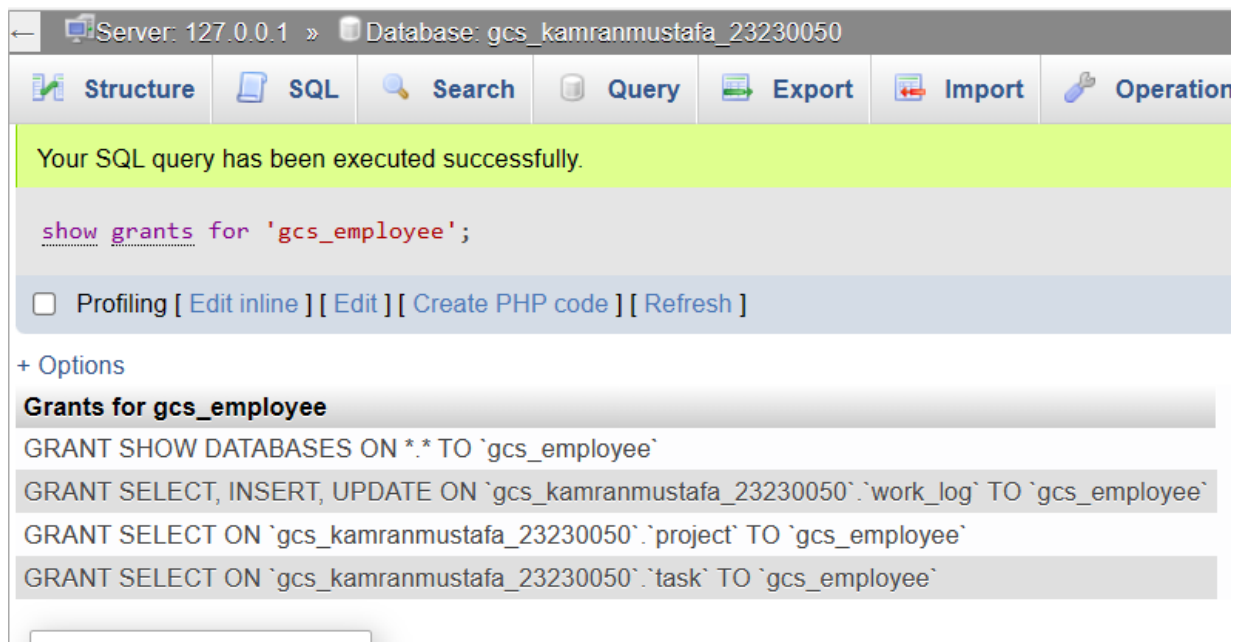


Figure 38 Grant for gcs_employee role



Figure 39 Grant for gcs_customer role

4.2 Create users and grant their role

For each role, at least 6 users were created.

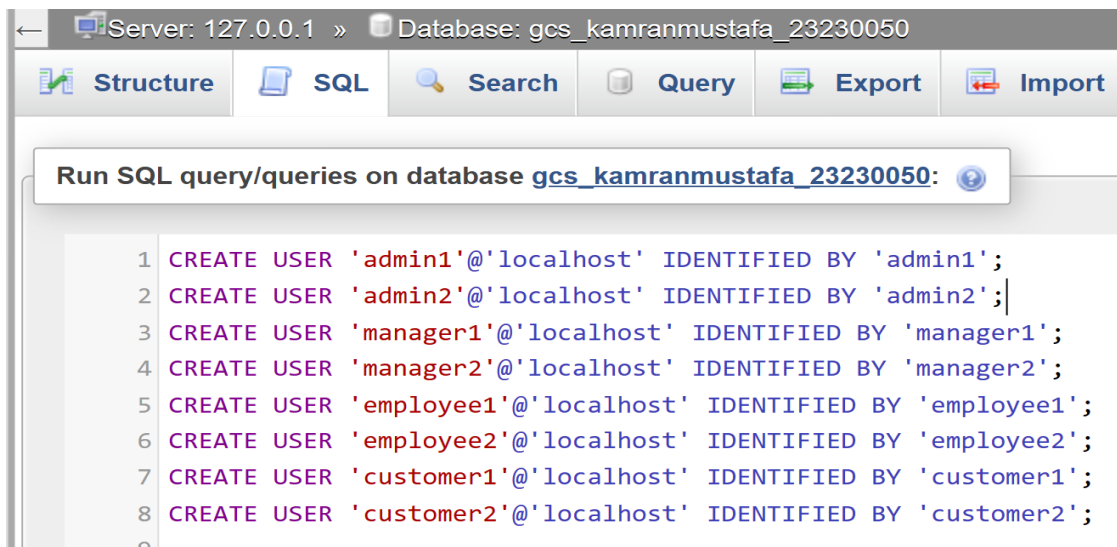


Figure 40 Creating the Users

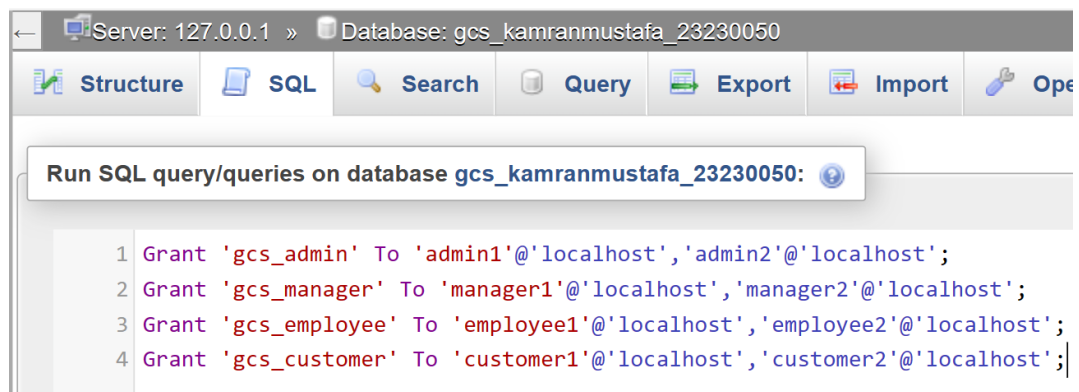


Figure 41 Granting roles to respective users.

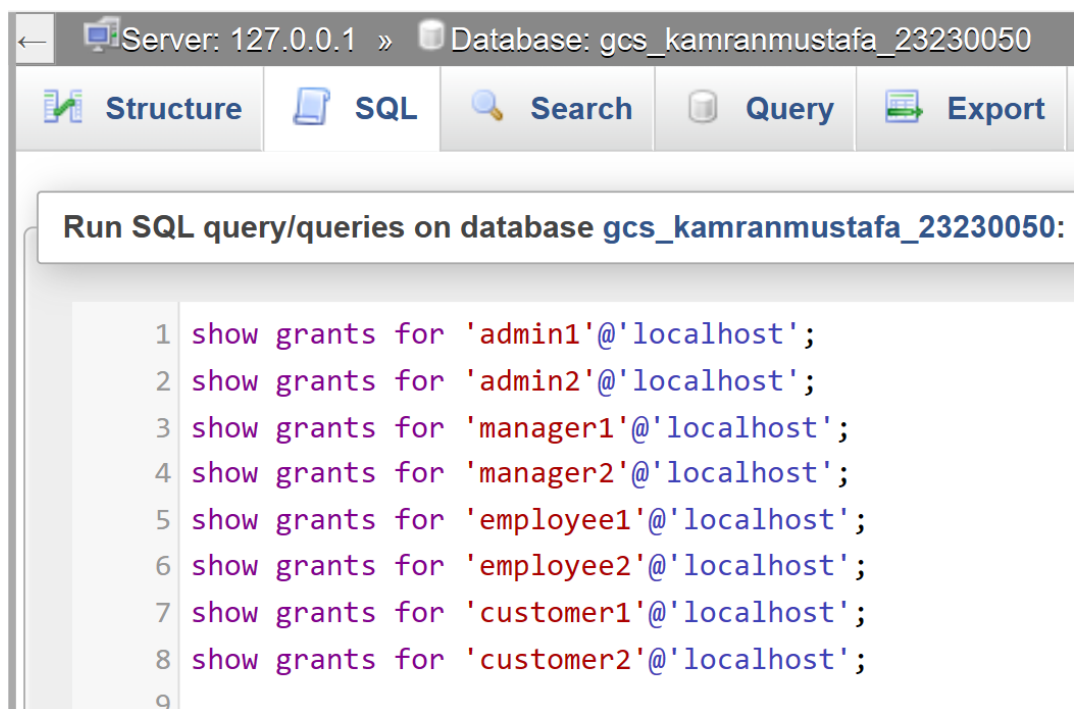


Figure 42 Showing roles grant to individual users

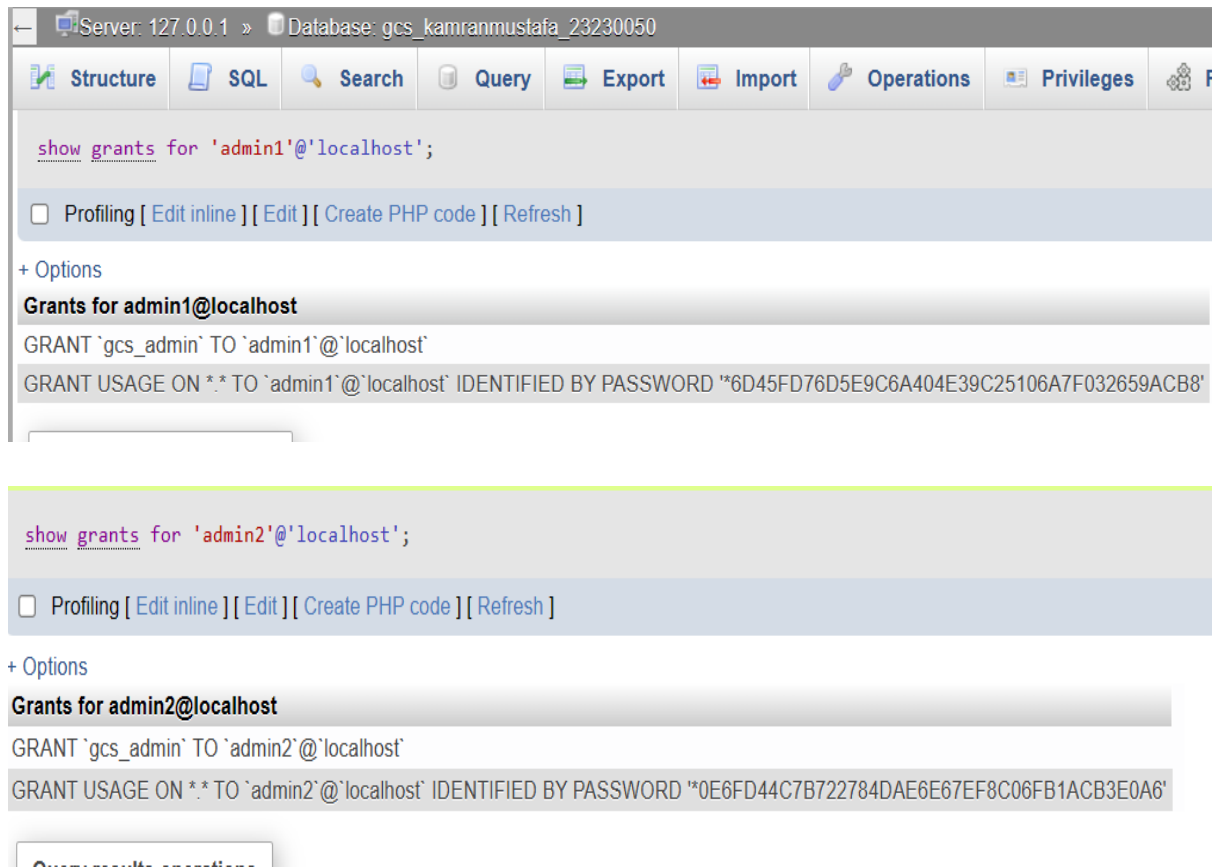


Figure 43 Role's grant to Admin1 and Admin2

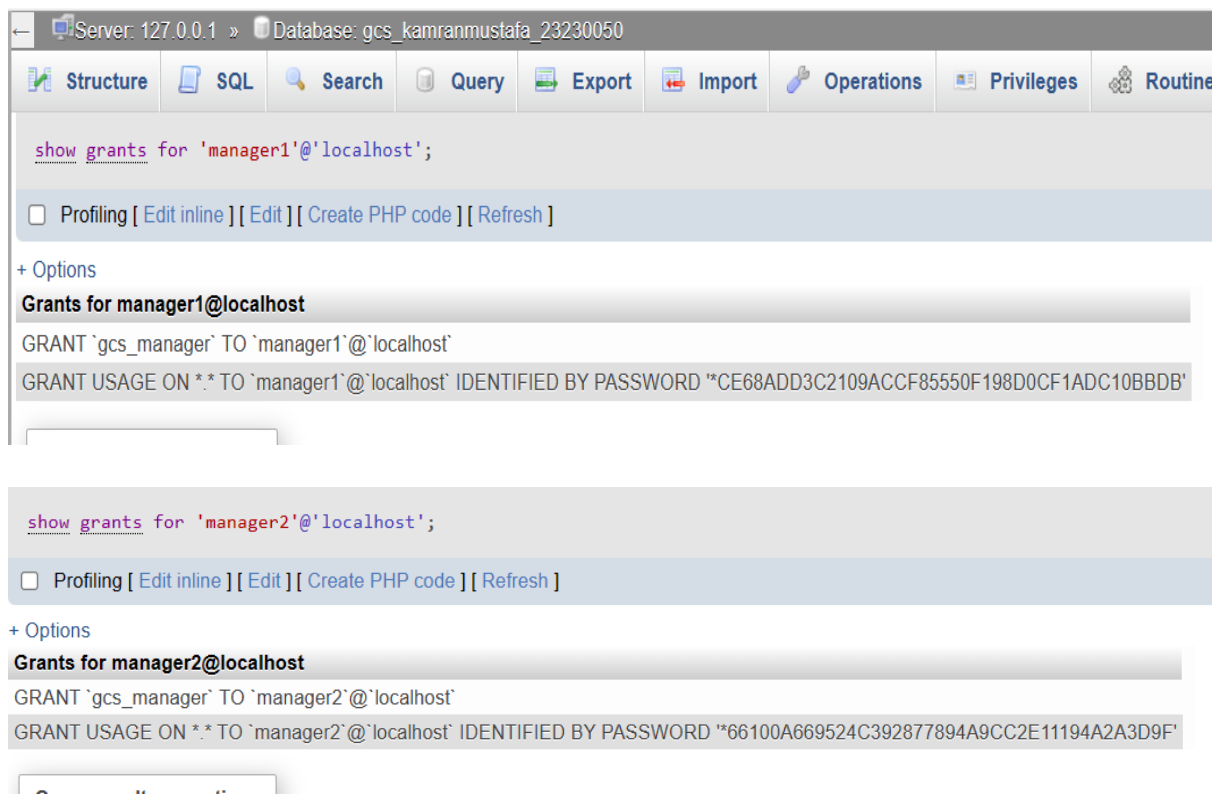


Figure 44 Role's grant to Manager1 and Manager2

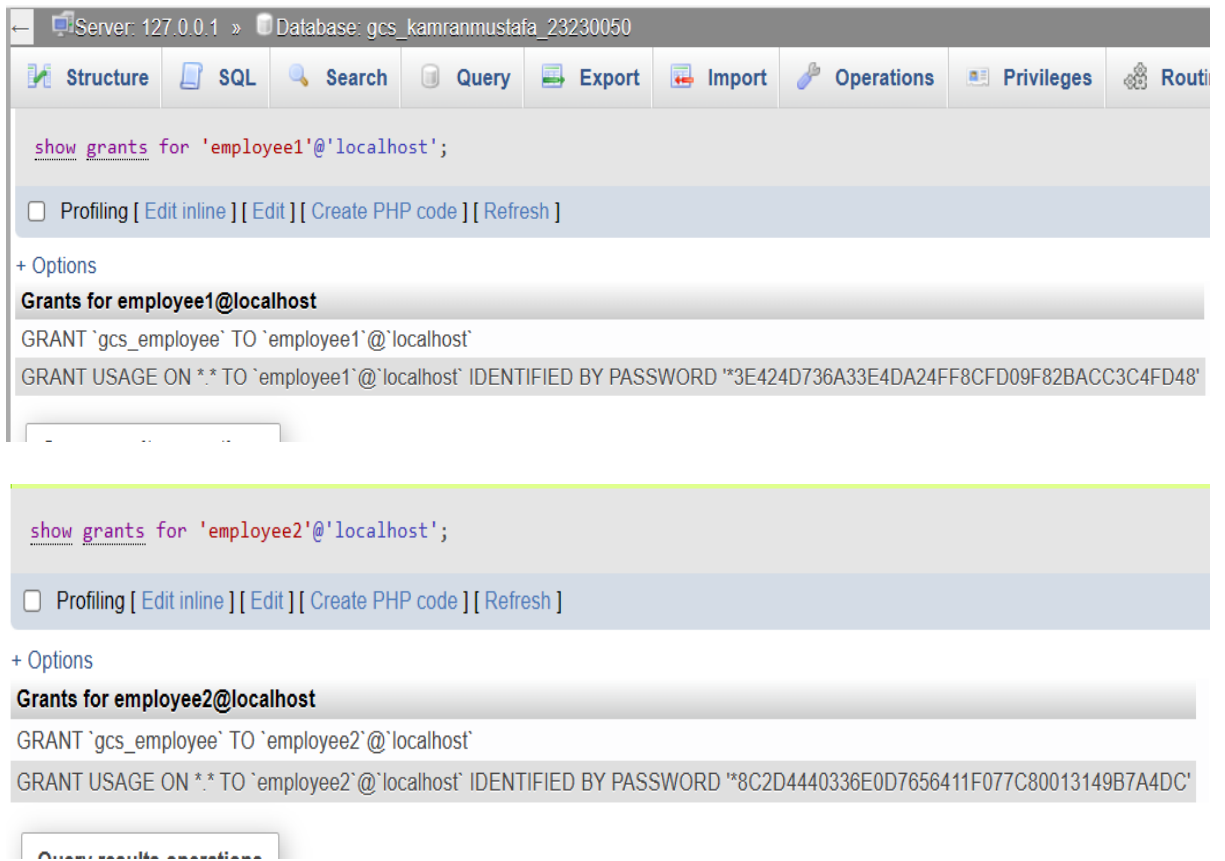


Figure 45 Role's grant to Employee1 and Employee2

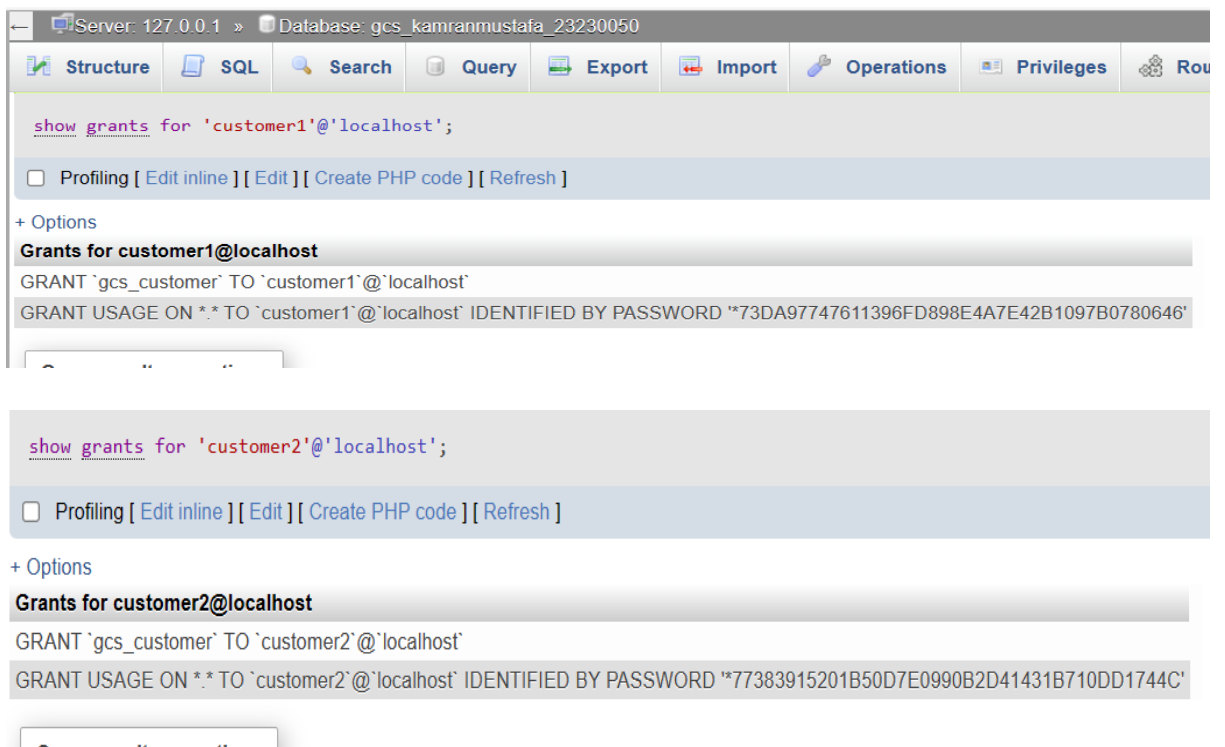


Figure 46 Role's grant to Customer1 and Customer2

5. Conclusion

A relational database is the best option for database technology in this case study because of its effective data organization into tables and its capacity to create associations between items using main and foreign keys. I've used Relational Schema Mapping and an Entity-Relationship Diagram (ERD) to organize the database, giving a clear picture of the entities and their relationships.

I then filled in the tables with a dataset that was tailored to the software company's requirements, which made testing easier and made it possible to create SQL queries for manipulating and retrieving data.

Finally, a security scenario is also implemented to create roles and assign privileges to users which ensures only authorized access is allowed in a database that maintains integrity and confidentiality.

6. References

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